GEK-83753C

DRIVE SYSTEM INSTRUCTIONS

MAINTENANCE MANUAL
For
CONTROL SIGNAL HIGHWAY
Driven By

MICROCOMPUTER BASED MASTER STATION

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

MAINTENANCE MANUAL

FOR

CONTROL SIGNAL HIGHWAY

Driven By

Microcomputer Based Master Station

(GEK-83753)

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

HOW TO USE THIS MANUAL

This manual is a user's guide to the maintenance of General Electric's Control Signal Highway which is driven by a Microcomputer Based Master Station.

We have written this manual to guide maintenance personnel in the installation, troubleshooting and maintenance of a Control Signal Highway.

We recommend that the inexperienced user read this manual from cover-to-cover to acquire an overview of the equipment maintenance requirements. Thereafter, the book will be a useful reference manual and should be kept near the equipment.

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--SECTION 1--
INTRODUCTION TO THE CONTROL SIGNAL HIGHWAY

General Information

This manual is a guide to the installation, operation, maintenance and troubleshooting of General Electric's Control Signal Highway (CSH) equipment. Section 1 describes CSH system structure, operating principals, application rules and message structure. Section 2 describes master and slave stations and how they operate. Section 3 includes important installation guidelines and installation checkout procedures for a newly installed CSH system. Section 4 describes how the master station firmware is structured to control the CSH. Troubleshooting procedures are aided by use of diagnostics described in Section 5.

We assume that a user of this manual has some knowledge and experience in electrical and electronics technology. Some prior experience with industrial control systems is also helpful.

Ralated Documents

Control Signal Highway Description DS3820SCM
Highway Test Module User Manual GEK-65151

CSH Architecture

The CSH is structured in the form of a transceive loop. As shown in Figure 1-1, a master station is located at the head of the loop which contains from one to fifteen slave stations.

The master station formats and transmits messages around the transceive loop. Signals returning from around the loop are received by the master station which checks the messages for errors and strips the variable information (data) from the message for use by the process computer.

Each slave station around the loop is addressable and contains a repeater which restores and retimes the message signal at the repeater output. This output becomes a delayed replica of the output of the master station at the head of the loop. The purpose of the slave station is to interpret the messages as they pass through. Thereafter, each slave station acts upon the messages addressed to it, ensuring that the control information is received and acted upon at its destination.

The function of the master station is to supply the content for the message format in prescribed bursts such that coordinated control of all devices connected to the slave station(s) is possible. Overall control of the master station is maintained by the microprocessor controller board DS3800HMPC.

There is an optional provision for a redundant loop. (See Figure 1-1). If the primary loop is broken, the system automatically selects the redundant loop. The system will continue to operate on the redundant loop even when the primary loop is restored. You can cause the system to return to primary loop operation by reinitializing the system or by reselecting the primary cable from the I/O diagnostic CRT described later in this document. If the primary loop is broken, it should be restored as soon as possible. If the redundant loop becomes broken while being used, the system will attempt to operate on the primary loop again, and it will do so provided the primary loop has been restored. However, if both the primary and redundant loops are open, the system will automatically switch back and forth between them while testing for a complete circuit. During this time, the process computer will detect the error and disengage automatic process control.

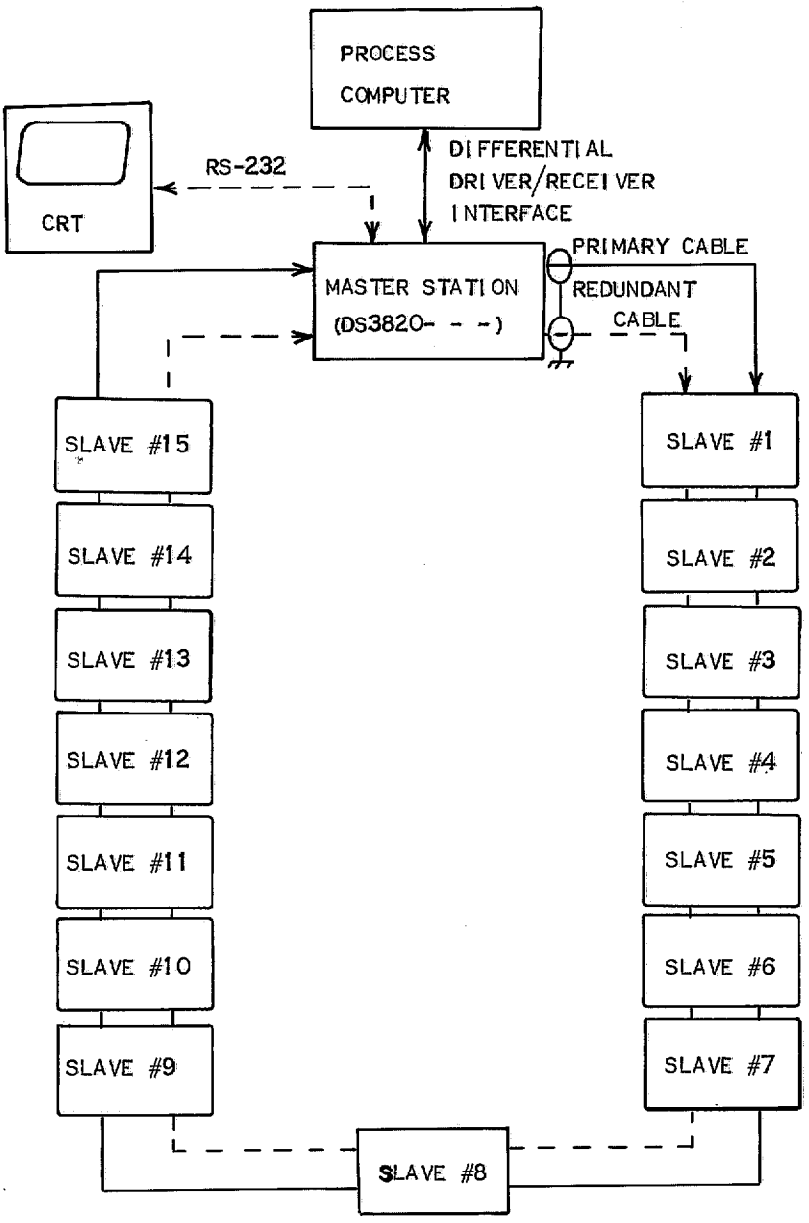


Figure 1-1
Basic CSH Loop

CSH Application Rules and Guidelines

1. Up to 15 addressable slave stations may exist along a CSH loop.

2. The strappable station addresses of all slave stations along a CSH loop must be unique and cannot be "0" Hex.

3. The maximum permissible loop length, L_{max}, in units of 1000 cable-feet is expressed by the following mathematical relationship:

$$L_{max} = 14.27 - 0.42S$$

"S" (above) is the number of slave stations along one loop.

4. The cable forming the CSH loop is Belden 8213 coaxial cable terminating at its ends in UG21BU connectors.

5. The Belden 8213 cable transitions to RG179 cable at each repeater through a coaxial connector plate provided in the PIU and slave cabinetry. The RG179 cable is provided as part of the equipment. The 8213 cable is provided by the customer.

6. The maximum recommended cable distance connecting three consecutive repeaters is 5000 feet.

7. The operating serial bit rate is 1.152 megahertz.

8. The bit error rate is less than 10^{-9}. The bit error rate is computed as following:

$$\text{Bit Error Rate (ERROR BITS/SEC}^2\text{)} = \frac{\text{ERRORS} \times 1.152 \times 10^6}{\text{SECONDS BETWEEN ERRORS}}$$

9. The CSH loop is electrically isolated at each repeater and will reject up to 8 volts peak to peak common mode out to 10 megahertz.

10. The isolators, cable connectors and the 8213 cable are all rated to withstand 500 VDC minimum.

11. CSH loops will operate reliably cross-temperature from 0 to 50°C. This means that one station along the loop can be in a 50°C environment while another station along the same loop can be in a 0°C environment.

12. Each CSH loop is grounded at a single point to the shield of the coaxial cables. The single point is at the head of the loop (PIU). This prevents error degradation from signals generated by ground loops. The ground wire is provided at the factory and connects the appropriate coaxial connector plate to the cabinet frame. The cabinet frame must be grounded to earth by the customer at installation time.

Process Interface Unit (PIU) Cabinet

The PIU cabinet contains the master station electronics module(s), power supplies, and in some cases slave I/O hardware.

Physical Description

PIU Cabinet Weight: 700 pounds (318 Kg)

Dimensions: Width - 36 inches(91.4 cm)
Depth - 36 inches(91.4 cm)
Height- 90 inches(228.6 cm)

Cooling: The PIU cabinet is cooled by convection.

Electrical Description

Input Power Requirements:

87 to 127 volts RMS. measured between input terminals ACLH and ACLL. (Located between TB2-2 and TB2-1).

Power Supply PS5 Outputs:

OUTPUT	CONNECTORS	PIN(S)	
+5 VDC,	40 Amps	JA thru JF	5
DCOM		JA thru JF	6,7
+15 VDC,	2 Amps	JA thru JF	10
-15 VDC,	2 Amps	JA thru JF	11
+28 VDC,	5 Amps	JA thru JF	8
ACOM		JA thru JF	12

CSH Protocol

Messages are contiguous and are identified from each other by a lack of modulation for a 1 1/2-bit period at the beginning of each message. The modulation is bi-polar and phase modulated as shown in Figure 1-4. Each bit time contains a pair of opposite-polarity pulses. The order of the pulse pair determines whether a binary one or zero is encoded. The encoded information is shown (in terms of message fields) just below the modulation sketch in Figure 1-4.

The SYNC field is the message separator. The S within the sync field is a single zero bit that allows phase lock onto the message.

The CMD BYTE contains the addressee of the message (i.e., CSH slave station) in 4 bits (X) followed by the intent of the message in the next 4 bits. The intent of the various message types is described in Table 1-1.

The INFO WORD field (2 bytes) carries the variable information within the message. Its contents can be addresses of devices served by the slave station. The contents can also be information to or from devices served by a slave station. When INFO WORD carries information from served devices, the field is sent as all zeros by the master station. Then this field is appropriately inverted by the addressed slave station according to information supplied, to that slave station, by the selected controlled device.

The extraction or insertion of information within the INFO WORD field occurs on-the-fly as the message passes through the repeater.

The CRC field checks the message content for errors over the span of the CMD BYTE field plus the INFO WORD field.

The FILL TIME field has no other function than to absorb time at least equal to the longest permitted propagation delay around a CSH loop.

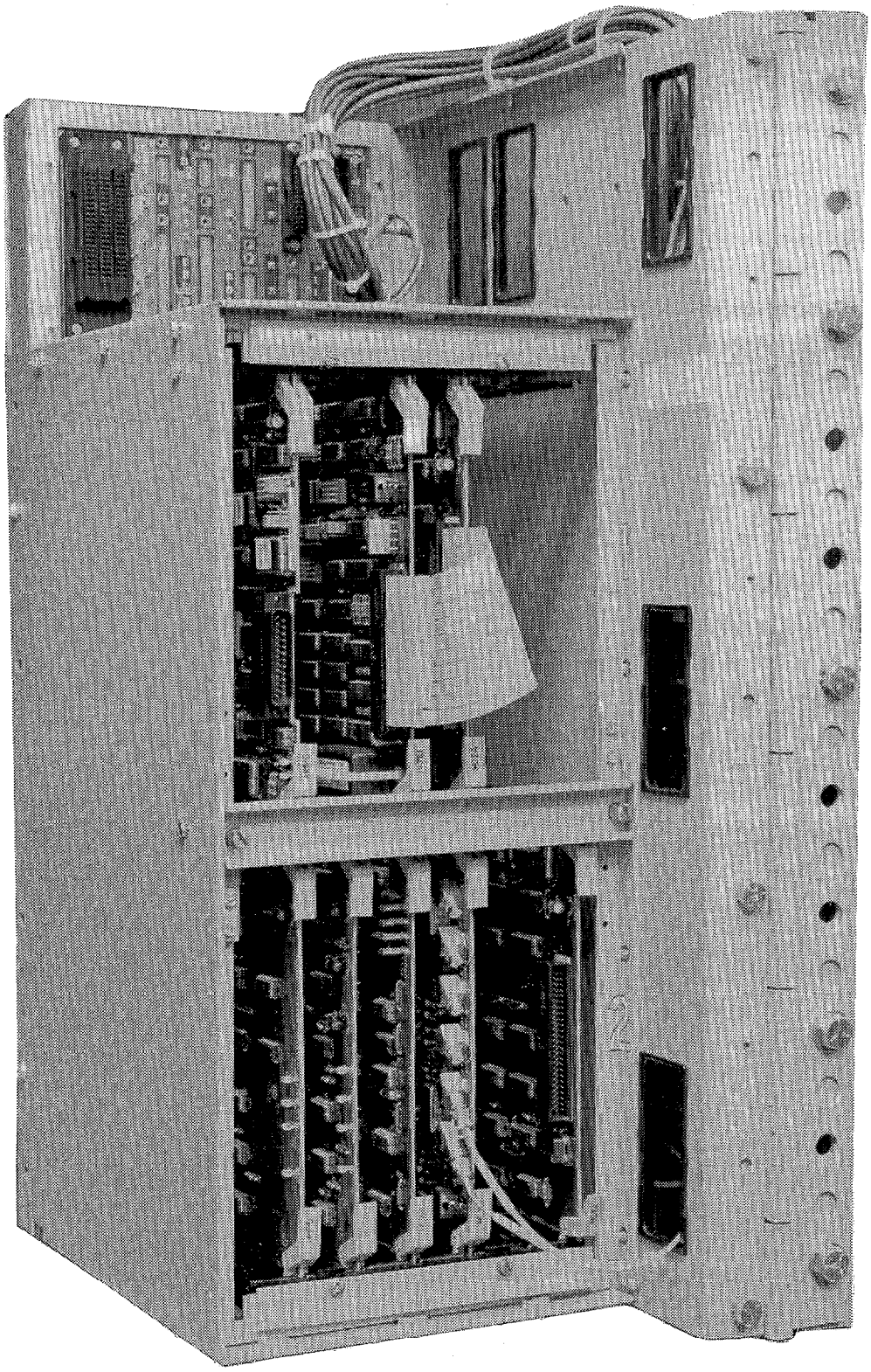


Figure 1-2
PIU Cabinet (DS3820SCM)

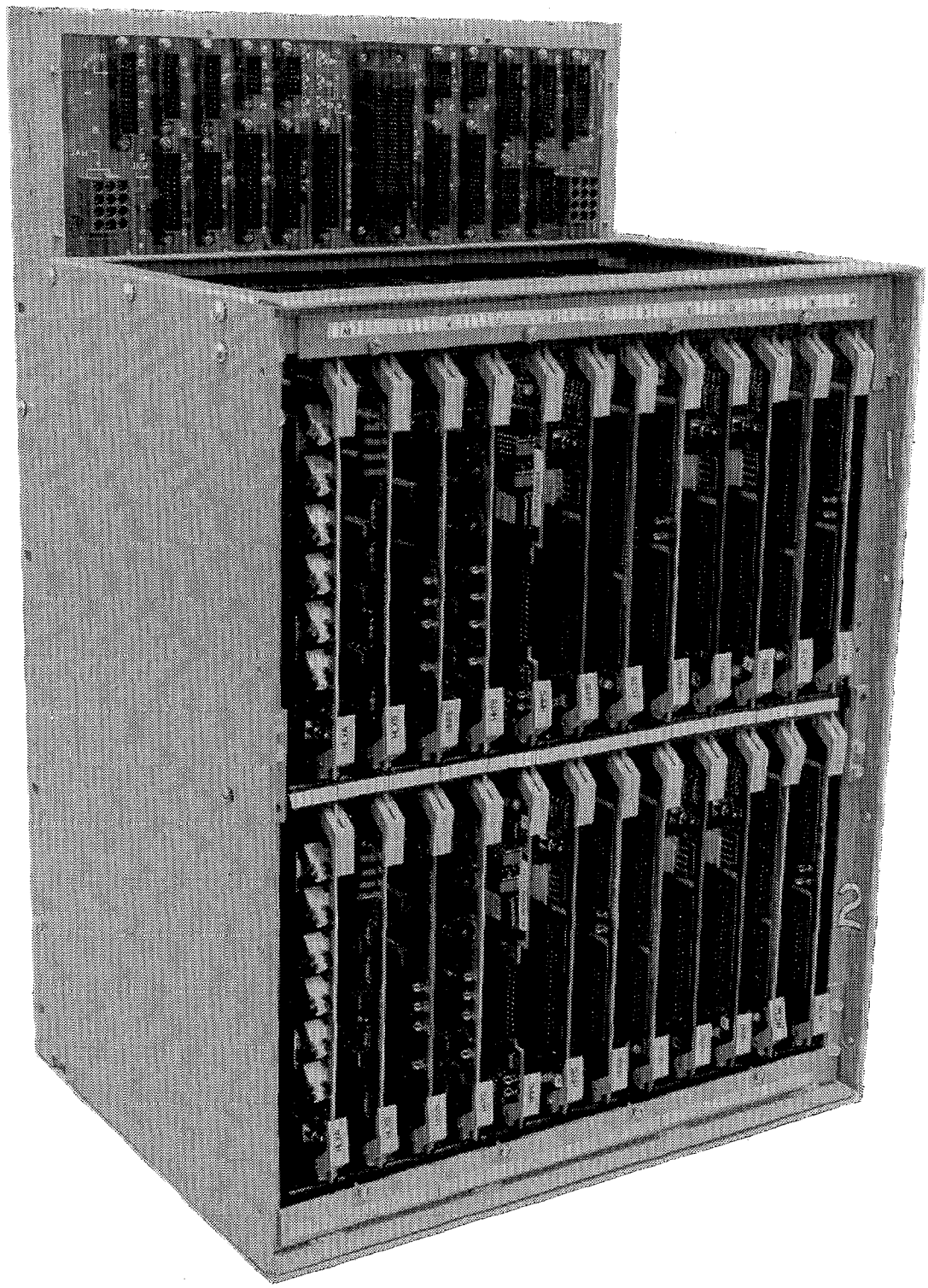
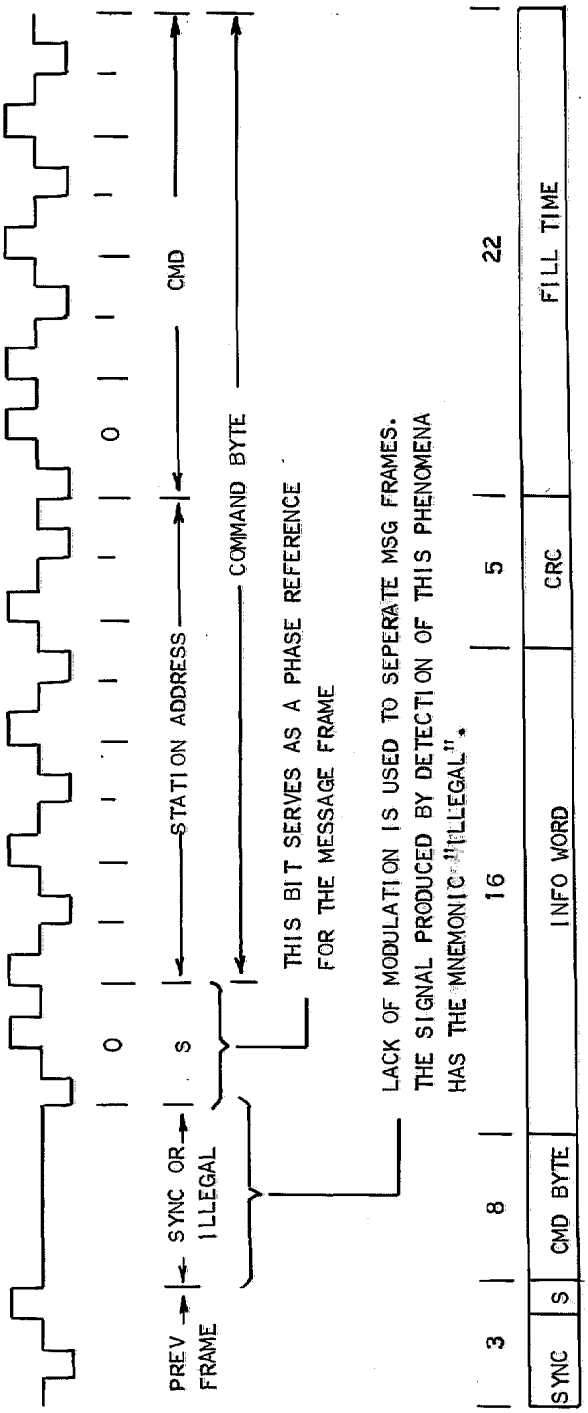


Figure 1-3
PIU Cabinet (DS3820MAB)



MSG FRAME BROKEN DOWN INTO FIELDS - BIT TIMES SHOWN ABOVE FIELDS.

FIGURE 1-4

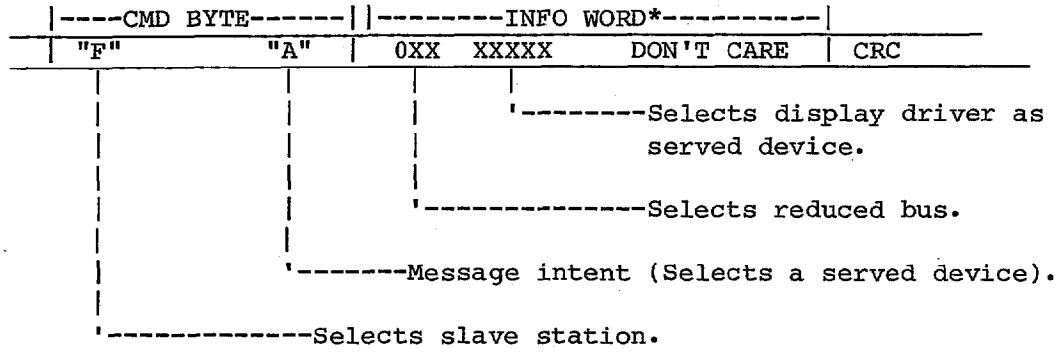
CSH MESSAGE FORMAT

Table 1-1
CSH Commands

MNEMONIC	CMD BYTE	MESSAGE INTENT
IDLE	00	IDLE: Occurs whenever there are no other messages to be transmitted.
SYNC	X0	SYNCHRONIZE: Message searches for the presence of station "X". The purpose is to establish loop viability during power-up or after link errors.
ACCA	XA	ACCEPT ADDRESS: Transmits LS byte of device address information to awake a device served by station "X" for the purpose of sending it control information in the next ACCD message.
ACCD	X9	ACCEPT DATA: Causes information to be sent to a previously awakened device served by station "X". After supplying this information to the served device, the slave station automatically increments the device address.
TLKA	XB	TALK ADDRESS: Transmits LS byte of device address information to awaken a device served by station "X" for purposes of acquiring information during the next "TLKD" message.
TLKD	XC	TALK DATA: Causes information to be acquired from a previously selected device served by station "X". After acquisition of this information from the selected device, the slave station automatically increments address to the next sequential device address.
INQ	XE	INQUIRY: This message is sent to station "X" to query it for the nature of its demand for special attention during a previous IDLE message.
INQR	XF	INQUIRE AND RESET: Same as INQ except that the reset line to I/O devices is activated.
ACK	X2	ACKNOWLEDGE: An acknowledgement to the slave station "X" that its needs for special attention will be honored.
ACCT	X8	ACCEPT TEST DATA: This message deposits test data within slave station "X".
TLKT	XD	TALK TEST DATA: This message retrieves the test data from station "X".
PAGE	X3	PAGE ADDRESS: This message transmits MS byte of device address information for the purpose of acquiring or sending data information.

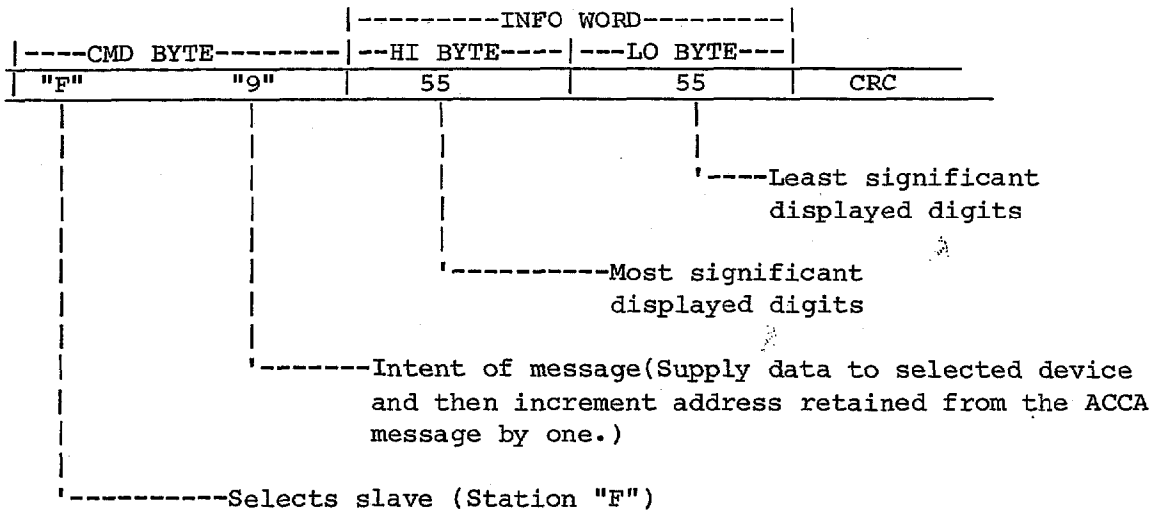
Information Transfer via CSH. During the following example of information transfer via CSH, refer to Figure 1-5. Consider the case of two types of messages listed in Table 1-1 which are sent to a remote device. They are an ACCA message and an ACCD message. The two messages are encoded and formatted as diagrammed below:

ACCA



* X denotes a variable binary bit

ACCD



Some devices (not illustrated) will accept only one byte (8-bits) of information although the message supplies 16 bits of information. Those devices are hard-wired to accept the information in the least significant byte and ignore the information in the most significant byte.

There are also data-supplying devices (not illustrated) capable of supplying either 8 or 16 bits of information into the INFO WORD field of a "TLKD" message. In this case, the eight bit device would supply the same information into the high byte and the low byte of INFO WORD field.

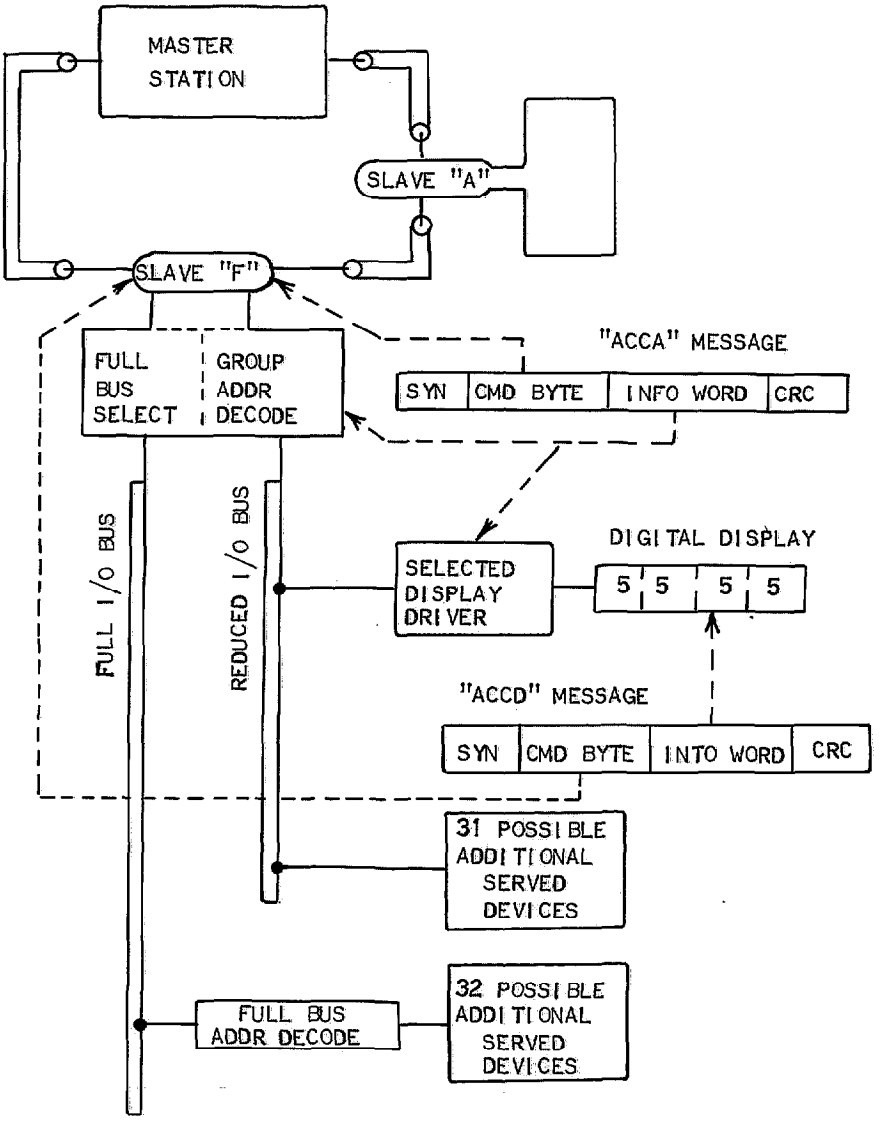


Figure 1-5.
CSH Information Transfer

During actual operation, messages are grouped into segments of data output messages and segments of data-gathering messages. The various message segments are run at fixed or variable time ticks as required by the application.

The CSH has the capability of accepting slave demands encoded into IDLE messages which require the manual intervention of the operator. Each station address has a dedicated bit assigned to it within the INFO WORD field of an IDLE message. This allows a demand to be asserted for special attention in these exceptional cases. The most significant bit is dedicated to station "F"; the next bit of lesser significance to station "E" and so forth.

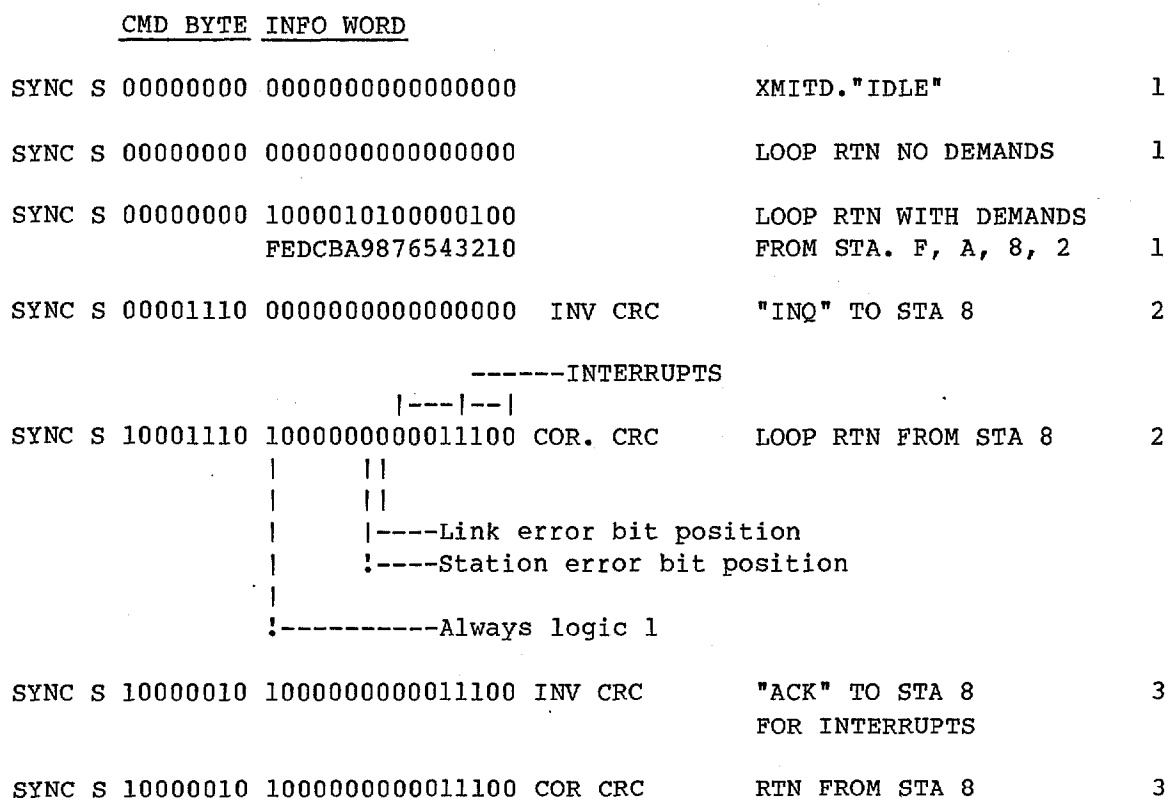


Figure 1-6
Remote Interrupt Servicing Messages

Figure 1-6 depicts a sequence of messages that accomplish the following:

1 Demands are discovered from multiple slave stations.

2 A query is sent to one of the slave stations to determine the nature of its demand.

3 An acknowledgement of the receipt of the query response is sent back to the slave station. This is required to turn off demands.

The underlined numbers (above) are a cross reference to the messages appearing in Figure 1-6.

Referring to message 2 of Figure 1-6, the link error bit, had it been present, would have denoted the detection of a faulty message during the transmission of the previous segment. The station error bit position, if set, would have denoted a lack of correct response by a served device when it was accessed by an "ACCD" or "TLKD" message during the previous segment. The least significant 8 bits are remote interrupts assigned on a requisition basis.

We have mentioned error reporting and error detection mechanisms throughout this section. The architecture of CSH is built upon a foundation of comprehensive error detection. Error detection is based on the 5-bit CRC field. It is transmitted inverted (incorrect) and is reinverted to be correct by the addressed slave. This constitutes a mechanism to detect a failed or an off-line slave. The CRC check also checks message integrity with the following error comprehension:

1) All single burst errors of 5 bits or less.

2) 95% of all single burst errors over 5 bits.

Additionally, there are message bit integrity checks performed at all repeaters that will flag illegal bit sequences. Finally, the master station compares a received "CMD BYTE" against its transmitted "CMD BYTE" counterpart. The net result is that it is virtually impossible to complete a faulty message transmission without detection of that faulty message.

Error detection is followed by message retransmission as a mechanism of error correction.

--SECTION 2--
DESCRIPTION AND OPERATION

Master Station Operation

A functional block diagram of the master station appears in figure 2-1. More complete master station diagrams appear in the sections of this document devoted to particular process computer interfaces.

The host interface provides a differential line driver/receiver interface plus the necessary control to transfer data, commands, status and interrupts between the process computer and master station. A complete host interface description is provided later in this document.

The 4K X 16 Bit RAM (HRMB) provides for buffering or storage of I/O data and control. This dual ported RAM board can be accessed from two separate buses by the master station controller (HMPC) or the process computer through the host interface. The software mapping of this RAM is provided in the master station software control section of this document.

The controller directs all operations performed by the master station. These operations include the following:

- Directs all I/O scheduling
- Provides CSH error and interrupt detection
- Controls all data transfers within the master station and on the CSH
- Provides CSH and master station diagnostics
- Provides RS232 interface to CRT for on and off line operation of I/O

The CSH transmit/receive functions are performed by four boards in the master station. These four boards are the HHRB, HHTB, HLXB, and HLXA. CSH messages are organized into a separate burst to output or input data, by the controller. The contents of each message includes two bytes of I/O information and one byte of command. This message is loaded into the XMIT register on the HHRB, which in turn is applied to the modulator (HLXA) along with the CRC and FILL characters generated on the HHTB. The serial bit stream is then applied to the CSH coax.

At CSH loop return, the HLXB breaks the demodulated message into command and information fields. The INFO word is read into the controller which in turn is written to the HRMB RAM. The CMDBYTE compare function checks the transmitted CMDBYTE against the received CMDBYTE. The CRC checker verifies the check result performed on the message stream. If an error is detected, a retry of the failed message is attempted. If the retry also fails, the process computer is notified of the error.

For a more complete description of master station operation, refer to the write-up on the particular master station in use (DS3820SCM, etc).

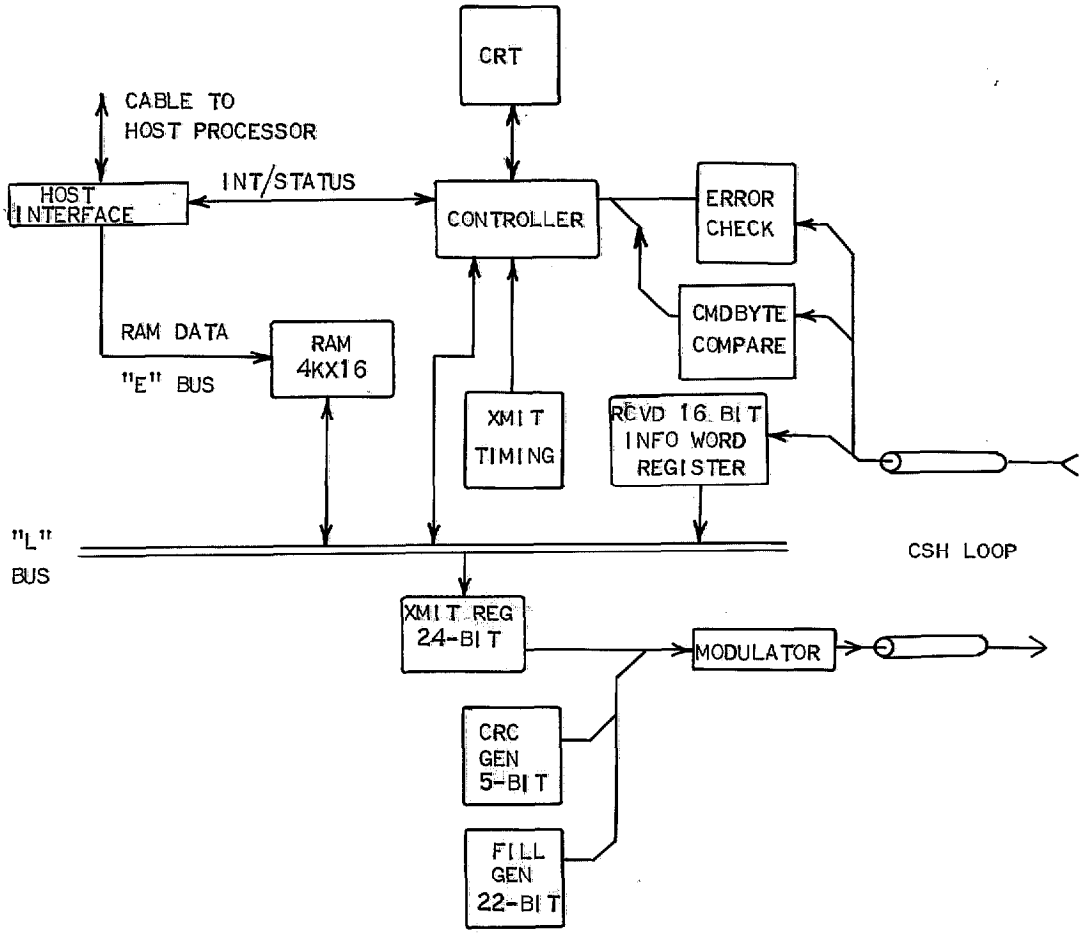


FIGURE 2-1
Block Diagram of Master Station

Slave Station Operation

The function of the CSH slave station is to recognize a message to it, and then to process the message according to its intent. The first byte, of the CSH message received at the slave, contains four bits of station address and four bits of message intent. If the slave is addressed, the slave controller captures the message intent and performs the message processing algorithm decoded. During this message processing, the next two bytes of serial information are extracted from the message. This data is converted to a parallel word and applied to address decoders or data drivers, depending upon the message intent.

The slave controller increments its I/O address register upon receipt of each "TLKD" or "ACCD" message address to the slave. This address register is also overwritten by each "TLKA" or "ACCA" message recognized by the slave station.

If a link error is detected by the slave, a message error is sensed by the slave controller and no data is sent to the I/O. Also a link error status bit is set within the controller and the master station is notified.

A bus integrity and parity check is also performed by the slave. If incorrect, a station error status bit will be set within the controller. This condition will also be flagged at the master station.

There is a safety mechanism employed at all slave stations. It is invoked when modulation of the CSH loop is missing beyond 6 milliseconds. Such an event causes the slave station to either hold all outputs at their last value or to reset them to zero. The choice is determined by backplane wiring on the slave and should only be made by qualified personnel.

At present the master station can interface to three types of slave stations.

1. DS3820TM2 - This slave is used to interface stick I/O with the reduced address bus or module I/O with the full address bus. This bus selection is controlled by address straps on the two HPBC boards. If one of the HPBC boards becomes active, it relays a "HOLD ACK" signal to the slave controller. The controller then enables the data lines along with defining control lines. An I/O card, recognizing the data lines being made available to it, will respond with a "DATA ACK" signal. When the DATA ACK signal is recognized by the controller, the data transfer is completed. The data transfer is between a register residing within the controller and an I/O device attached to the I/O data lines. A functional block diagram and picture of this slave are shown in Figure 2-2 and 2-3.

2. DS3820MMI - This slave station functions similarly to the DS3820TM2 slave station. The difference being that all CSH slave and I/O interface boards are all self-contained within the same electronics modules. This type design is applied when slave station mounting is required within operator stations. A functional block diagram and physical layout of this slave module are shown in Figures 2-4 and 2-5.

3. DS3820CSM - This slave station functions as a CSH interface to the reduced address bus, the full address bus and the dual ported RAM board HRMB. To provide for this increase in I/O addressing, a PAGE address is required in the message sequence to this slave. The following PAGE addresses have been assigned to each mode of addressing:

   AO HEX - Reduced address bus (STICK I/O)
   CO HEX - Full address bus (BUFFER-CUSTOM I/O)
   60 to 7FH - Interface to HRMB (8K bytes) which provides 32 pages of 256 bytes each.

The HRMB interface provides the data interface required betwen the CSH and the DLC Plus Sequencer. The Sequencer, which is controlled by an HMPC microcomputer board, provides distributed processing capability. A functional block diagram and physical layout of this slave module are shown in Figures 2-6 and 2-7.

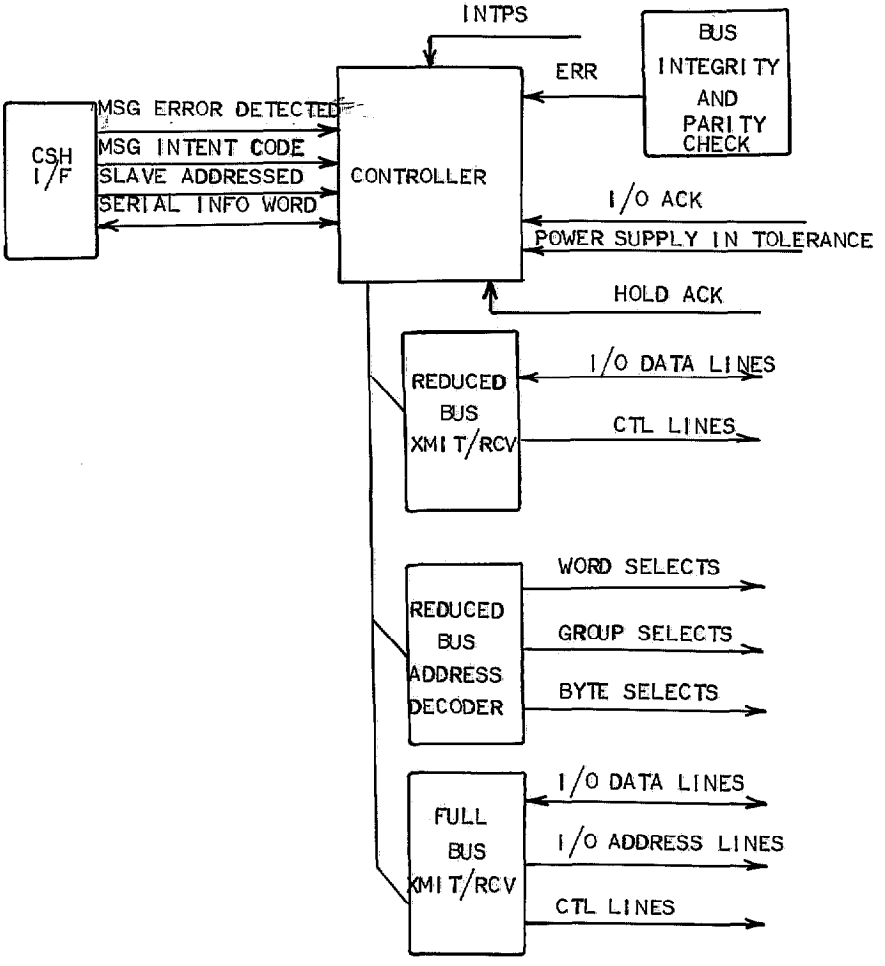


Figure 2-2
Block Diagram DS3820TM2 Slave Station.

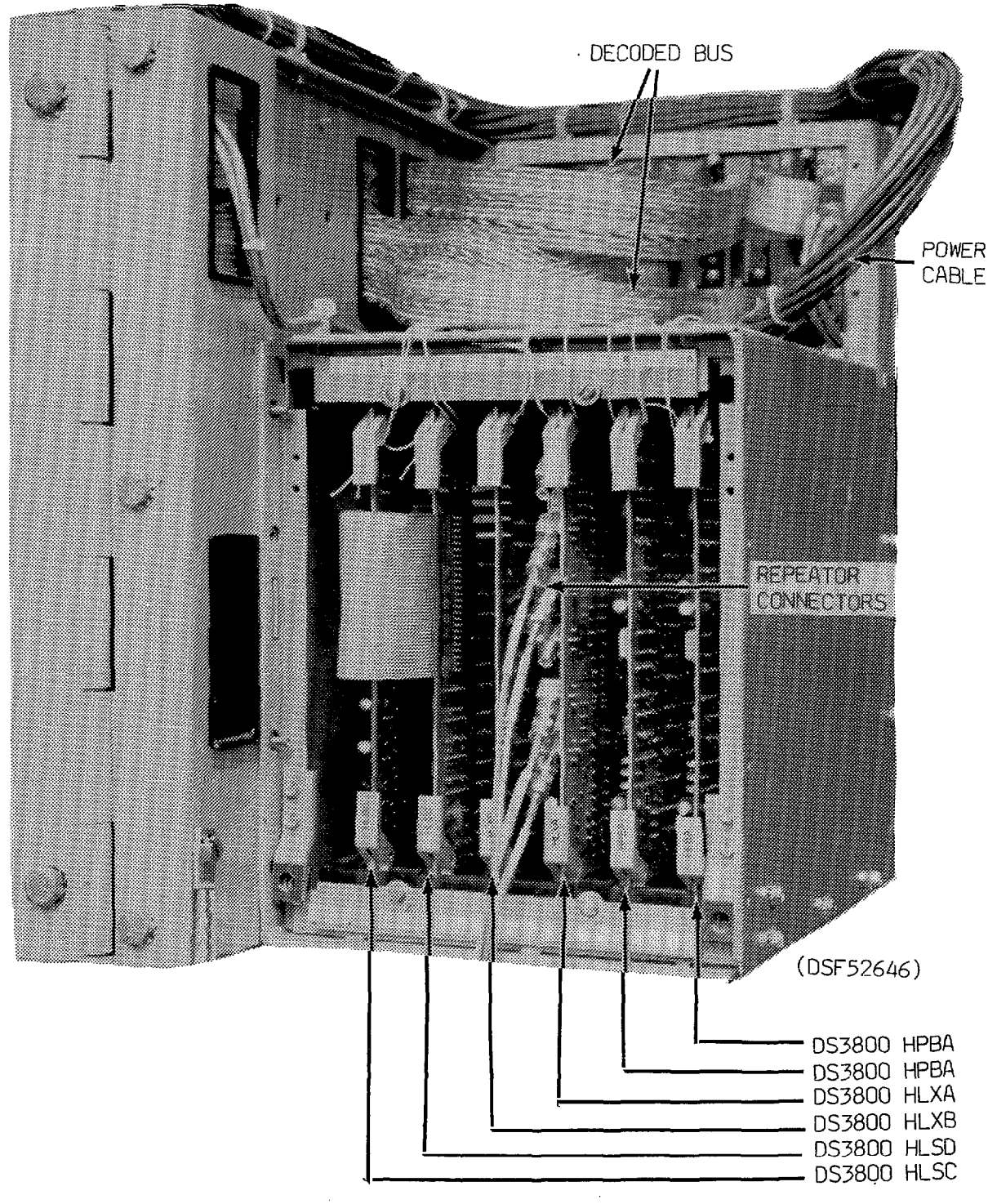


Figure 2-3
DS3820TM2 Slave Station

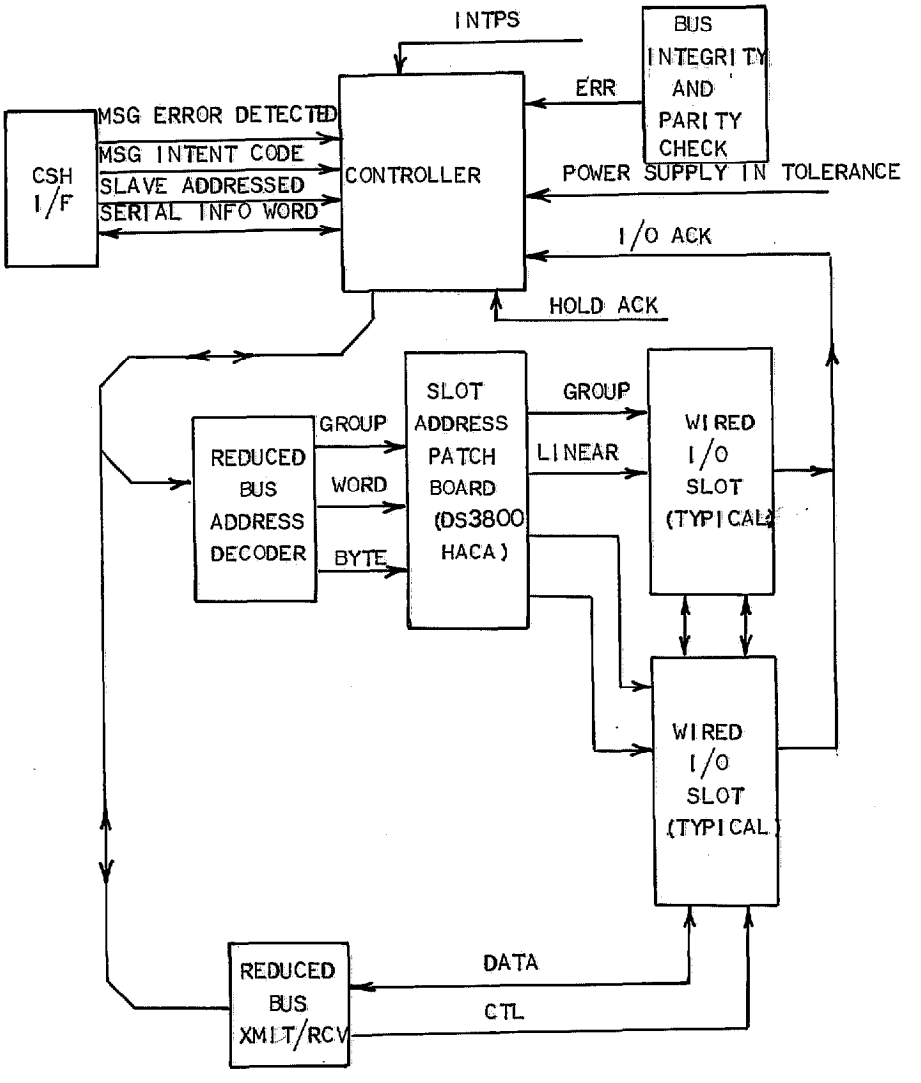
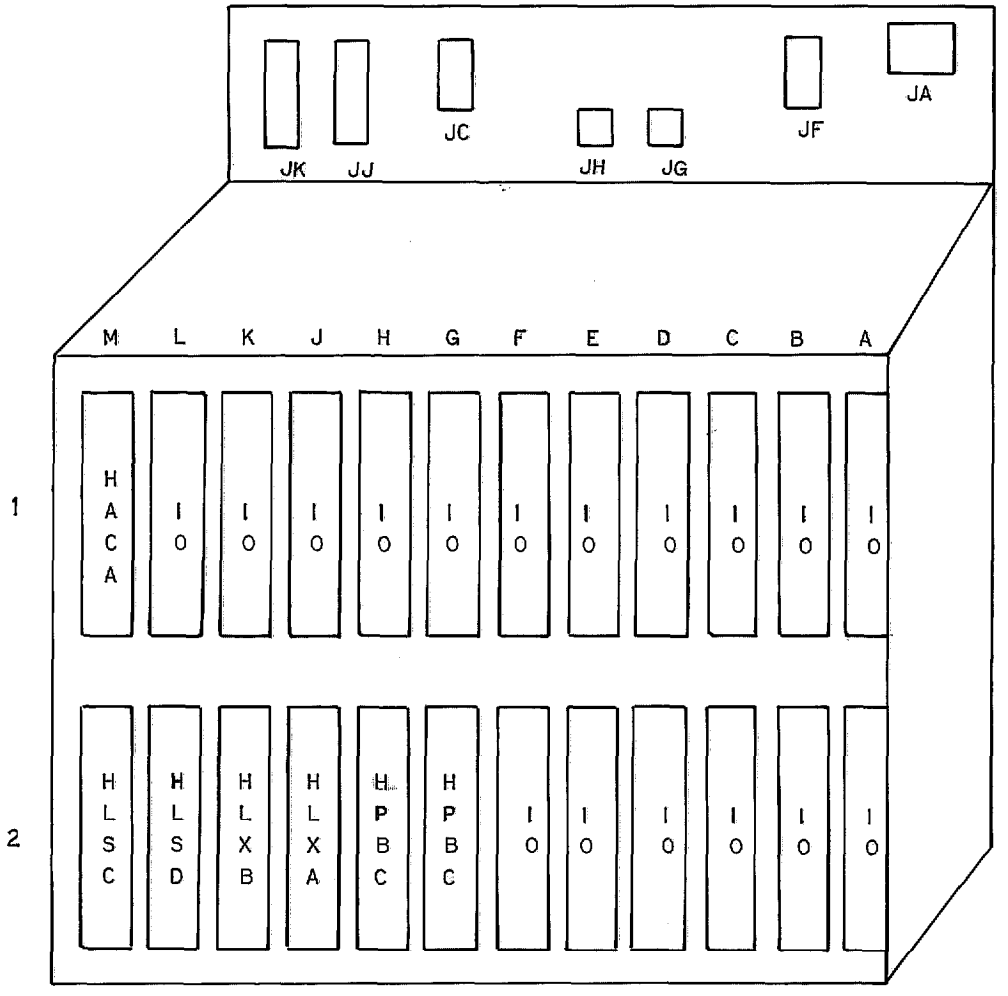


Figure 2-4
Block Diagram DS3820MMI Slave Station



JJ AND JF ORIGINATE DECODED I/O BUS
JK AND JC ORIGINATE UNDECODED I/O BUS
JH,JG CABLE BETWEEN THEM BRINGS DIRECT DATA OUT OF I/O SLOT 1A TO CSH INTERRUPT INPUTS AT SLOT 2L
JA POWER SUPPLY CONNECTION

Figure 2-5
DS3820MMI Physical Layout

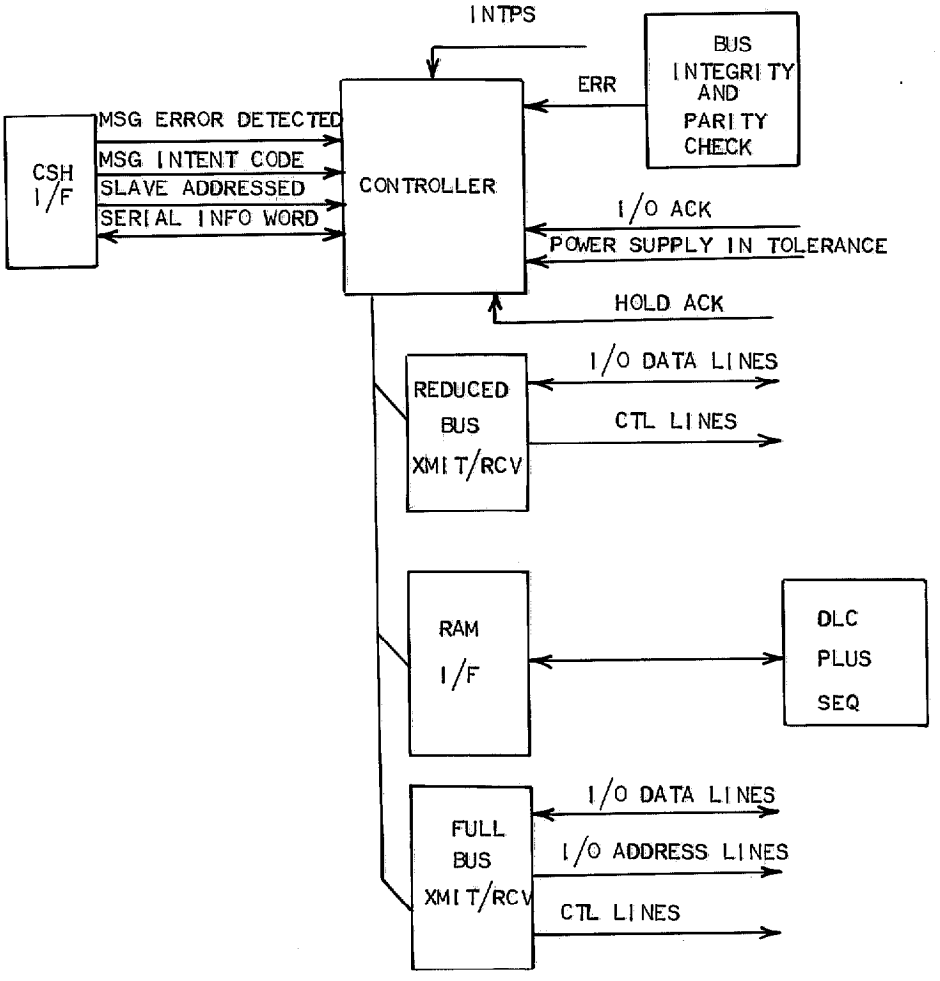
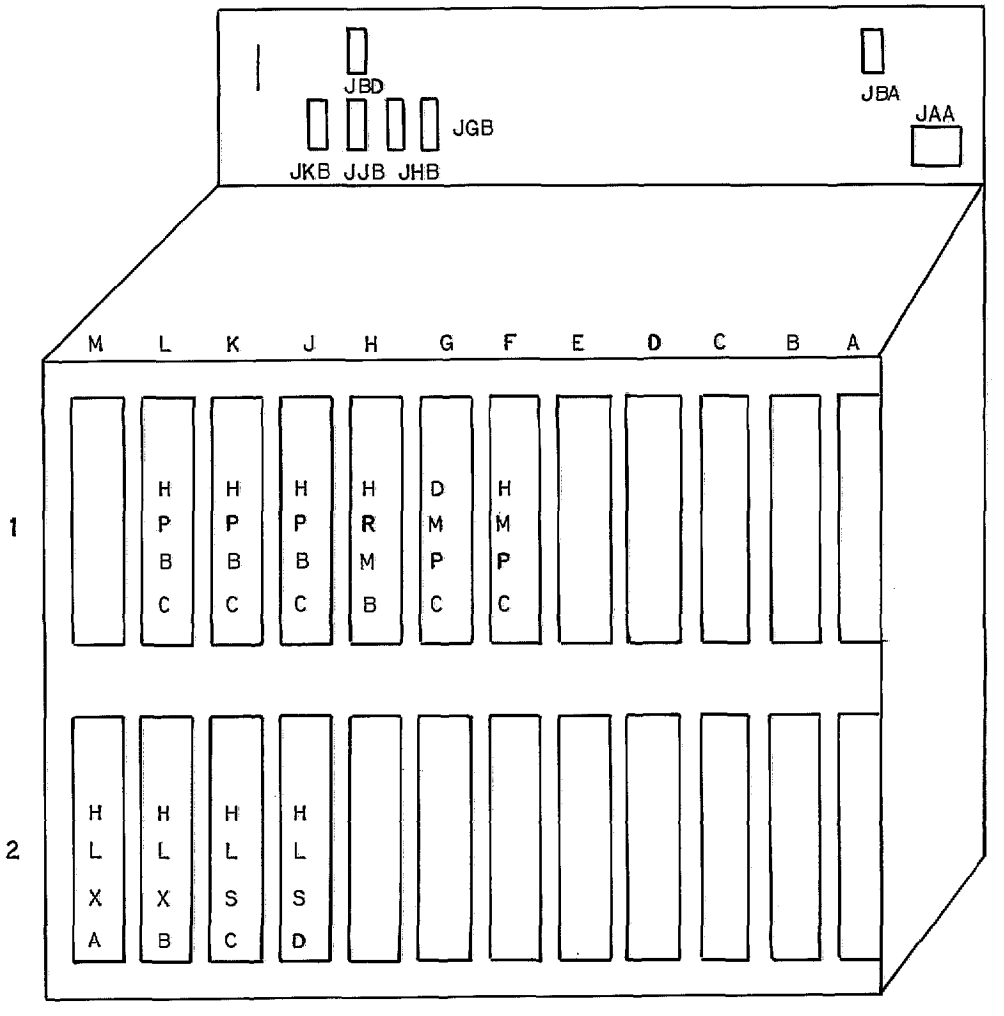


Figure 2-6
Block DIagram DS3820CSM SLave Station



JJB + JGB ORIGINATE DECODED I/O BUS
JHB + JKB ORIGINATE UNDECODED I/O BUS
JBA + JDB CABLE BETWEEN THEM BRINGS DIRECT DATA OUT OF I/O SLOT 1A TO CSH INTERRUPT INPUTS AT SLOT 2L
JAA POWER SUPPLY CONNECTION

Figure 2-7
DS3820CSM Physical Layout

Master Station/Honeywell 4500 Interface

A complete block diagram of this master station appears in Figure 2-8. Physical layout, board content and cabling are shown in Figure 2-9.

The interface between the master station (DS3820SCM) and the H4500 process computer consists of the Honeywell MPIS (Microprocessor based Process Interface Subsystem) board and the DS3800HCIA board.

The MPIS controller board contains the necessary hardware/firmware to create a parallel data link between the H4500 GENIE BUS and the line driver/receiver interface to the HCIA. The 1K of firmware on the MPIS decodes instructions, enables downline loading of H4500 memory to 3K of on-board RAM, enables up-line loading of this RAM and performs self diagnostics. The 3K of RAM is downline loaded, each time the H4500 is initialized, with the micro-code required to operate the H4500-to-master station interface. Refer to Honeywell documentation for a complete description of MPIS decoded instructions and board operation.

The microcode downline loaded at initialize time performs the following functions:

1. Verifies the MPIS to HRMB(RAM) and HMPC to HRMB data interfaces and flags all errors detected by master station diagnostics.

2. Initializes CSH I/O system parameters and control word tables.

3. Controls the DMA transfers of all data between the H4500 and the HRMB.

4. Detects H4500 stall.

5. Transfers master station status to the H4500.

6. Enables MPIS to HRMB hardware checkout described in diagnostic section of this document.

The MPIS board provides a differential line driver/receiver user interface consisting of 8 output data bits (A0-A7), 4 output command bits (C4-C7), 8 input data bits (B0-B7) and 4 input status bits (C0-C3). The interface cable is 7.5 meters long and is made of standard computer grade twisted pair cable.

The HCIA block diagram is shown in Figure 2-10. Four command bits are received by the HCIA from the MPIS. Bit C7 is used directly as a master station hardware reset. Bits C4 thru C6 are decoded on the HCIA to generate the 7 commands described in TABLE 2-1. Bits C0 and C1 are the only input status bits used. C0 gives an indication to the MPIS that the HRMB has completed a read or write operation. C1 indicates that the HMPC (Master Station) has stalled and that the process I/O is not valid.

Bits A0-A7 are used to output one byte of address or data from the MPIS to the HCIA data or address register. Two bytes of data transfer are required to load either the HCIA address or data registers. Bits B0 through B7 are used to input data only from the HRMB, one byte at a time.

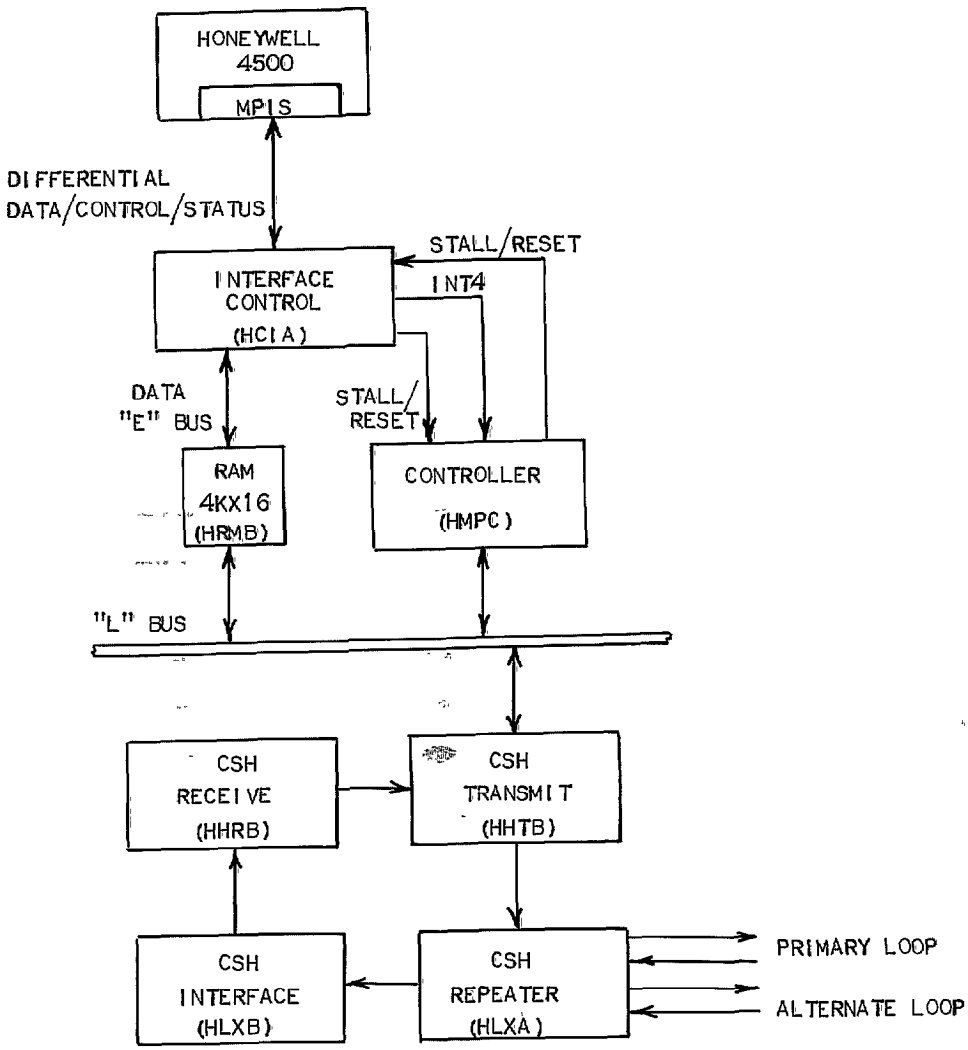
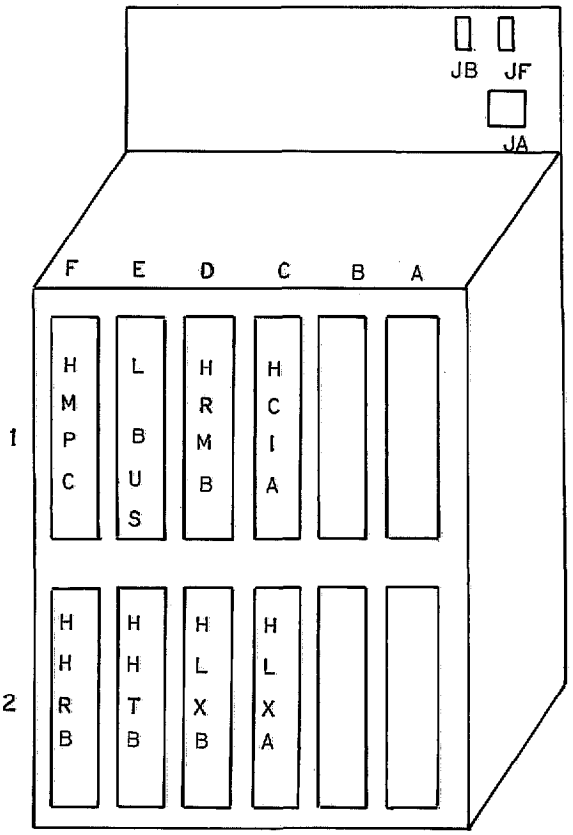
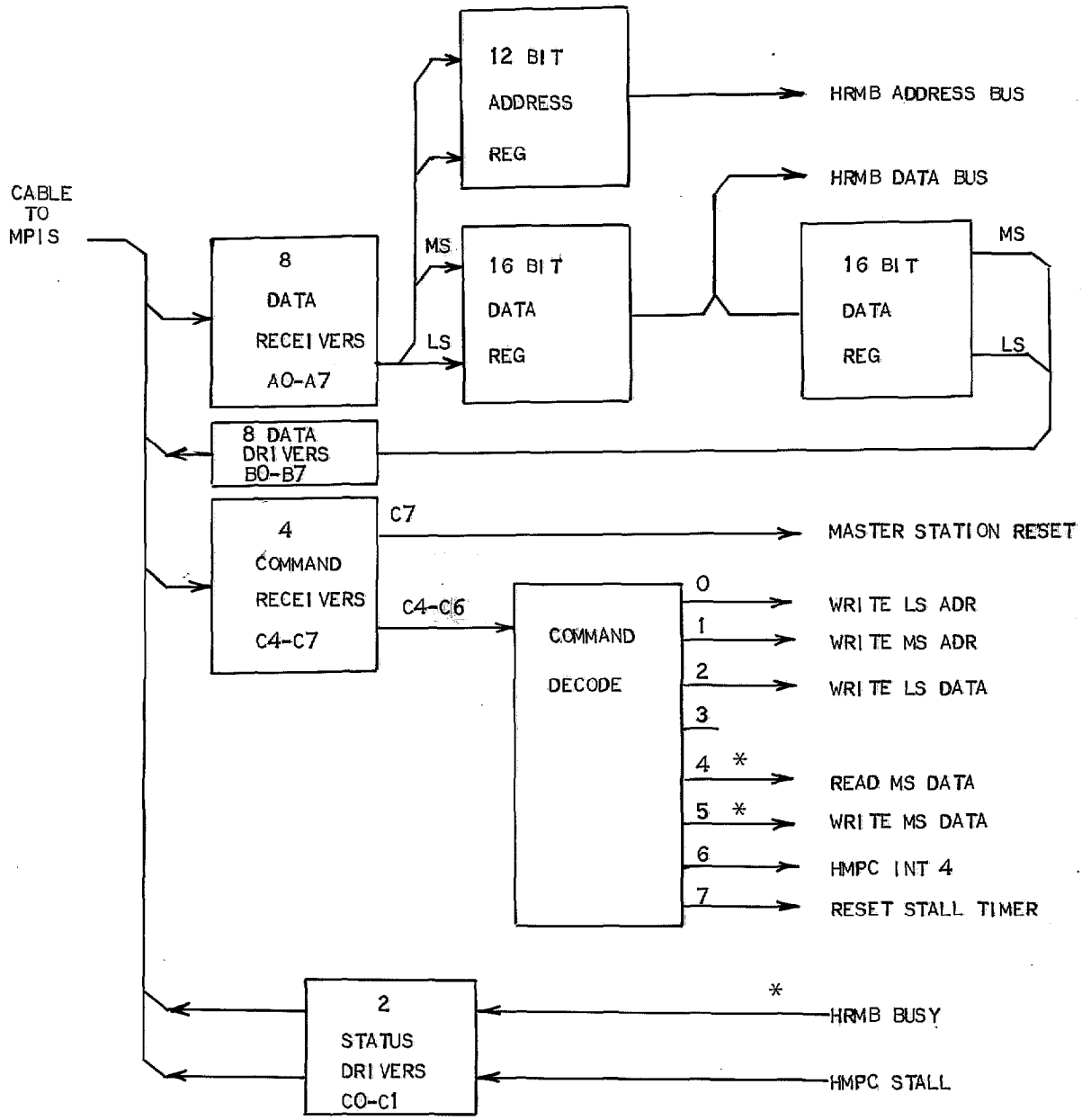


Figure 2-8
Master Station - Honeywell
DS38320SCM



JA POWER SUPPLY CONNECTION
JB+JF MASTER STATION/H4500 INTERFACE CABLE
HLXA CSH COAXIAL CABLES
HMPC RS232 CONNECTOR TO TERMINAL
HCIA/HRMB 50 PIN RIBBON CABLE "E" BUS

Figure 2-9
DS3820SCM Physical Layout



* HRMB BUSY INDICATES COMPLETION OF RAM R/W STEPS ADDRESS REG BY 1

Figure 2-10
DS3800HCIA Block Diagram

Table 2-1
MPIS/HCIA Commands

C6 - C4	COMMAND DESCRIPTION
0	Loads data receivers A0-A7 into HRMB address register LS byte.
1	Loads data receivers A0-A3 into HRMB address register MS nibble.
2	Loads data receivers A0-A7 into HRMB data register LS byte.
3	Not decoded but can be used to send HRMB data register LS byte thru data drivers B0-B7. After reading MS byte with command 4, the next 4 command will read in LS byte.
4	Reads HRMB data reg MS byte thru data drivers B0-B7. BUSY bit C0 indicates completion of write. Steps address register count by 1.
5	Loads data receivers A0-A7 into HRMB data register MS byte. BUSY bit C0 indicates completion of write. Steps address register count by 1.
6	Generates an interrupt to the HMPC - INT4.
7	Reset MPIS deadman timer. The MPIS must output this command every 200 msec or less, to avoid a stall condition to the master station.

NOTE: THE HCIA ONLY ACKNOWLEDGES COMMAND CHANGES.

Master Station/DEC*Unibus Interface

A block diagram of the master station module DS3820MAB is shown in Figure 2-11. This module can generate two CSH loops with each loop capable of interfacing to four separate computers thru dual port RAM boards (HRMB). The physical layout of the master station module is shown in Figure 2-12. This module was designed as a general purpose module, that can be configured with board selections, to meet multiple system interface requirements.

The interface between the master station and the process computer consists of a DEC DR11-BD interface and the two Drive Systems boards DS3800HCIB and HDRA.

The DR11-BD consists of a DR11-B general purpose DMA UNIBUS link and an M5947 differential driver/receiver board. This interface is capable of transferring blocks of data, under computer control, between computer memory and the differential data interface. The data interface is a 16 bit word interface with appropriate control and status as shown in Figure 2-13. A functional description of the data interface is given with the HCIB board description. Refer to DEC documentation for a complete description of DR11-BD operation.

The HDRA board provides 16 differential line drivers and receivers that are used to interface data words between the DR11-BD and the HCIB interface board. The interface cable connects to the HDRA by two card edge connectors JA and JB and connects to the HCIB board by card edge connector JA.

The HCIB block diagram is shown in Figure 2-13. The interface cable control and status signals connect directly to the HCIB board by the card edge connector. These signals are described in Table 2-2.

The HCIB interface to the DR11-BD is initialized and controlled from the process computer through a driver program. The driver program sets up the DR11-B word count, bus address, data buffer and status registers and then initiates an interface operation by setting the 'GO' bit in the status register.

The DR11-B word count register is initially loaded with the number of words to be transferred. After all DMA transfers have been completed, word count overflow occurs and stops word transfers.

The bus address register specifies the next sequential computer memory location to be read or written by DMA. This register will increment after each word transfer to allow for the block transfer of sequential memory locations.

The data buffer register is used to transfer all data words between the process computer and the master station interface. It is loaded with the starting HRMB RAM address during an initial read or write interface setup. The address word is then loaded into the HRMB select (Bits C-F) and address (Bits 0-B) registers on the HCIB when the 'GO' bit is set in the DR11-B status register.

*DIGITAL EQUIPMENT CORPORATION

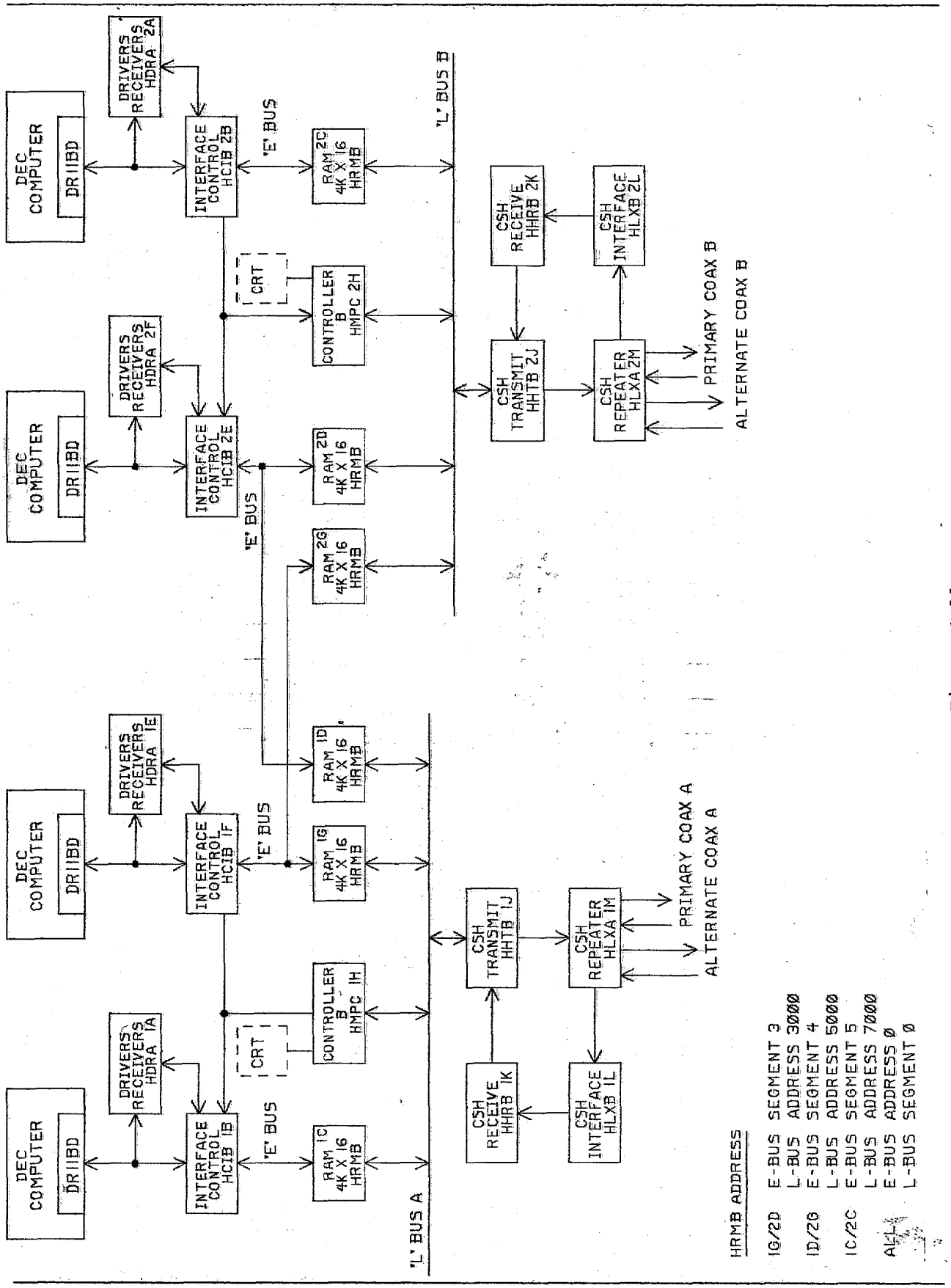
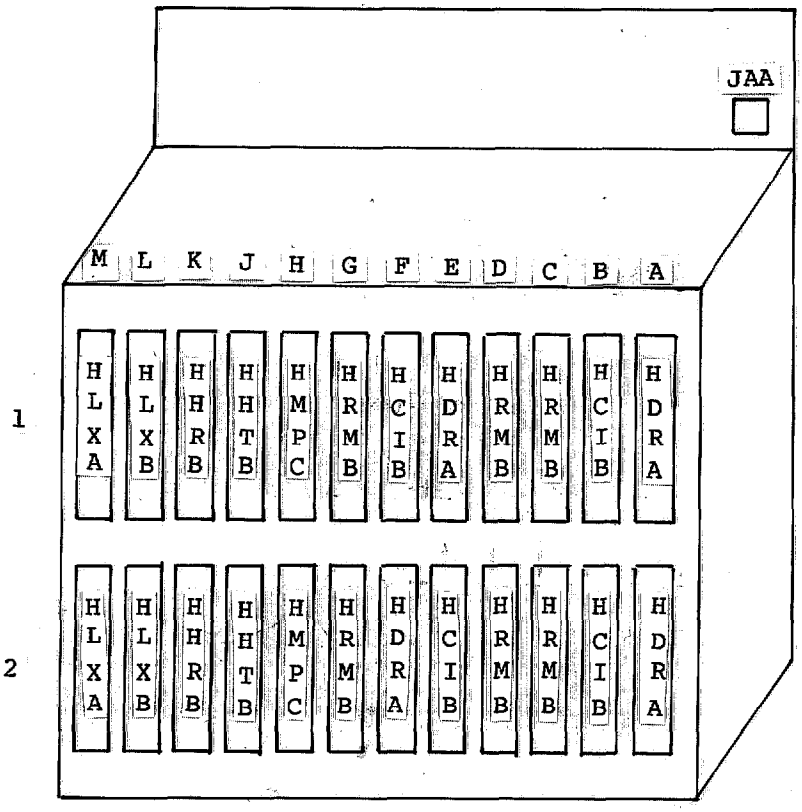


Figure 2-11
Master Station - DEC Unibus
DS3820MAB



JAA	POWER SUPPLY CONNECTION
HDRA	DATA INTERFACE CONNECTORS JA AND JB (34 PIN)
HCIB	CONTROL INTERFACE CONNECTOR JA 26 PIN
HCIB/HRMB	50 PIN RIBBON CABLE "E" BUS
HLXA	CSH COAXIAL CABLES
HMPC	RS232 CONNECTOR TO TERMINAL

Figure 2-12
DS3820MAB Physical Layout

The status register is used to issue commands to and receive status from the master station. Refer to DEC documentation and Table 2-2 for the description and use of the status register.

The normal sequence of reading or writing to the HRMB RAM board from the process computer starts with DR11-B register setup. When the HCIB board detects the 'GO' bit set, the HRMB address is loaded into the HCIB from the interface data output bits DAT00-15 OUT. After the HRMB address is loaded, the HCIB will control the interface in the read or write mode depending on the decoding of FNCT 1,2,3 bits. When the DR11-B word counter overflows, all data word transfers are stopped.

The HMPC controller interrupt (INT4) is generated in the same manner as an HRMB read or write except that the DR11-B word counter is set equal to one and the HRMB address word, that is loaded from the DR11-B data buffer, is not used. Refer to the user's guide section of this document for the function of the HMPC interrupt.

The driver program resident in the process computer performs the following functions:

1. Verifies the process computer-to-HRMB and HMPC-to-HRMB data interfaces and flags all errors detected by master station diagnostics.

2. Initializes CSH I/O system parameters and control word tables. The process computer which interfaces to HRMB address '00' will control the CSH loop associated with that HRMB.

3. Initializes DMA block data transfers between the DEC UNIBUS and CSH master station.

4. Decodes master station status.

5. Provides for DR11-BD to HRMB hardware diagnostic checkout described in diagnostic section of this document.

Table 2-2
DR11-BD Interface Signals

INPUT SIGNALS	
DAT00-15 IN	16 bits of data input from master station to DR11-BD.
C1 CONTROL	Controls the direction of DR11-BD data transfer. Logic 0 = output data to master station. Logic 1 = input data from master station.
CYCLE REQUEST A	Master station request the use of the UNIBUS to perform a DMA word transfer.
ATTN	Forces an error condition in the DR11-BD if the HMPC controller stalls. (Disabled on DS3820MAB)
DSTAT A,B,C	These three bits are loaded directly into the DR11-BD device status register. A - CSH coax in use 0 = primary 1 = alternate B = CSH system error has occurred C = CSH hardware interrupt has occurred
OUTPUT SIGNALS	
DAT00-15 OUT	16 bits of data output from DR11-BD to master station.
INITIALIZE	Hardware reset to master station.
FNCT 1,2,3	These three DR11-BD status register bits are decoded by the HCIB for interface setup. 0 = Write to HRMB 1 = Read from HRMB 2 = Reset HMPC 3 = 6 Not Used 7 = HMPC interrupt 4
GO	Used by the HCIB to indicate that a new operation (read, reset, write, interrupt) has been initiated by the process computer.
READY	Used by the HCIB to enable ATTN.
END CYCLE	Indicates that a UNIBUS DMA cycle has completed.

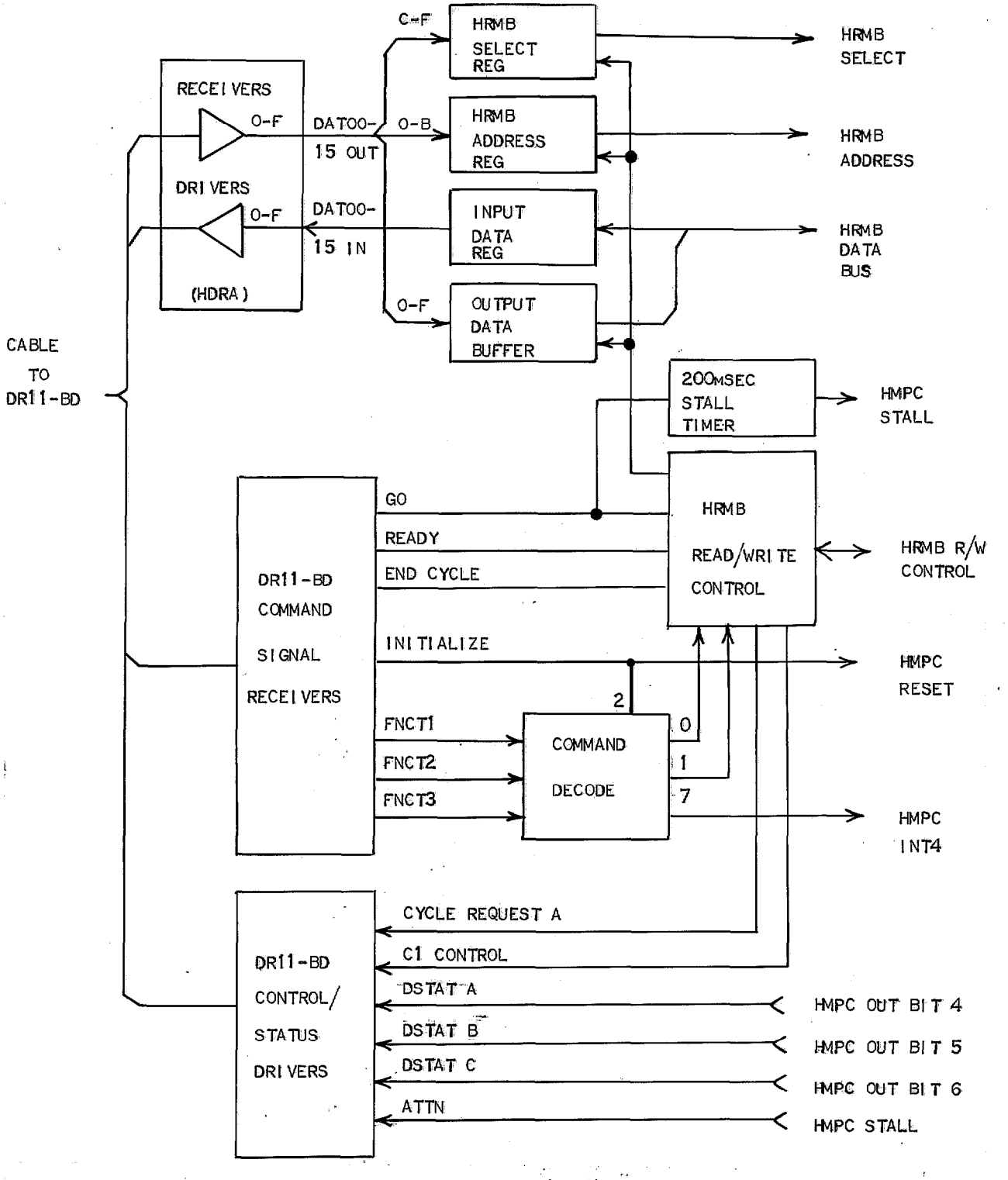


Figure 2-13
DS3800 HCIB BLock Diagram

Master Station/MODCOMP* Interface

The block diagram of the master station module DS3820MAB, connected to a DEC computer, is shown in Figure 2-11. The interface to a MODCOMP computer is identical; a 5852 parallel interface takes the place of the DEC DR11BD.

A 5852 interface consists of two MODCOMP cards and associated cables. The cards are electronically and mechanically identical. Dip switches on each card are used to configure the cards to perform two different functions: transmit/control and receive.

A brief overview of how the 5852 operates in conjunction with the DS3820MAB will follow, however, an in-depth understanding may only be obtained by studying the following MODCOMP documents:

MODCOMP NO.	TITLE
111-A00094-001	5852 GE/DSD Parallel Interface Diagnostic
509-A00187-001	MODCOMP GE/DSD - CSH Master Station Interface
517-A00085	Assembly Drawing - 5852 Parallel Interface
502-A00047	Elementary Diagram (Logic Diagram)
225-200217-102	Technical Manual Model 4805-2 (Especially the chapters in the MODCOMP I/O bus).
GE No.	Title
278A2182AG	Design Specification for Special Purpose MODCOMP Interface

The normal mode of operation for the 5852 is a continuous, "Free Running" mode. Data structures are built which contain commands instructing the interface what operation to perform as well as the address of where the data is to be moved to/from. The data structures are chained back on themselves; that is, after finishing all transfers instructed by the data structures, the interface starts over again.

This method of interface control is extremely CPU-efficient; the CPU is never involved in data or control transfers once the interface is initially started up.

*Modular Computer Systems, Inc.

--SECTION 3--
INSTALLATION INSTRUCTIONS

****
* *
* WARNING *
* *
****

DO NOT PERFORM THE FOLLOWING PROCEDURE UNLESS ALL DRIVES ARE DISABLED FROM RESPONDING TO CSH-TRANSMITTED MESSAGES. OTHERWISE, PERSONNEL NEARBY THE PROCESS COULD BE INJURED AND EQUIPMENT COULD BE DAMAGED.

Initial Installation

The following procedures should be performed during the initial installation phase of any CSH loop to insure the integrity of the system.

1. Terminate the customer cables by installing connectors per assembly instructions given in Figure 3-1.

2. When installing CSH cabling in open cable trays or conduit, avoid routing coaxial cables through a cable tray also containing AC power wiring.

3. Connect customer cables to form the required CSH loops according to the configuration shown in Figure 3-2. If it becomes necessary, for any reason, to lengthen the cable, cable-splicing connectors must be isolated from ground to prevent possible ground loops.

4. Assume that installation variables, i.e., jumpers on various PWBs, are installed correctly. However, you can become familiar with these variables by referring to Figures 3-3 through 3-14 and the accompanying tables.

5. Slave stations are normally installed with I/O connections intact. Assume that the I/O is already intact. Be sure to ground cabinets to earth ground at the cabinet ground lug.

6. Turn on power at each slave station. Also, switch each repeater into the CSH loop via SW1, located on the repeater (DS3800 HLXA). Verify "Run" LED is lit (HLXA).

7. Verify that the LVSH and LVSL signals at TB2-5 and TB2-6 of the slave power supplies are at logic 1 and logic 0 respectively. If not, the supply voltages are out of tolerance and will require adjustment.

8. Remove power to master station for the CSH loop under test. Remove the PAC interface board HCI*. Turn power on to the master station.

9. Verify IMOK LED is lit on the HMPC board. If not lit, reset the HMPC by jumpering TP1 to TP2 (HMPC). IF still not lit, take corrective action as described in User's Guide section (initialization) of this document.

10. Verify that the LNK LED is lit on the HHRB board and that the ACTV LED is lit on the HHTB board. The PCMC LED (HHTB) indicates which CSH coax is selected (on = primary coax). IF the LNK LED (HHRB) is lit, verify that the other CSH coax loop is closed, if applicable.

11. If the CSH loop is open (LNK LED off), walk the loop in the direction of serial data flow. As long as the loop is closed, the LKACT LED on the HLSC board will be lit. IF the LKACT LED is off, this indicates the existence of a break in the coax. If an oscilloscope is available, the HLXA board TP1 and TP2 can be checked for valid data as described in Table 3-1.

12. Verify that the following master station primary loop LEDS are lit.

    RUN - HLXA
    LNK - HHRB
    ACTV - HHTB
    PCMC - HHTB
    IMOK - HMPC

13. Switch to the alternate Coax using the CRT terminal command "ACOA". The PCMC LED should go out but remaining LEDS in step 12 should remain lit, if applicable

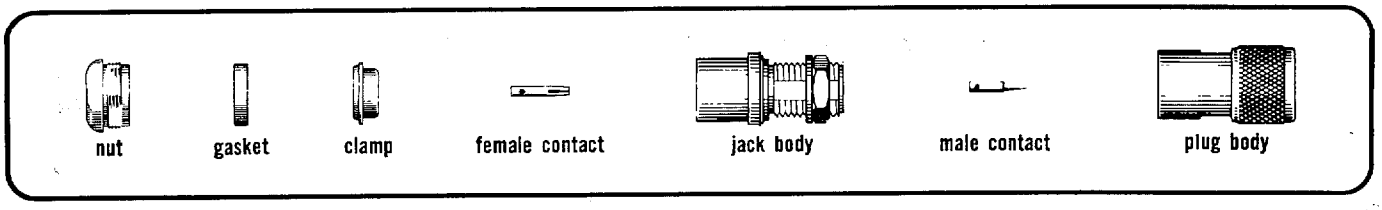
14. The function codes listed in Table 4-4 can be called from the CRT terminal to completely check out the CSH loop.

15. Remove power to the master station and insert HCI* board. Turn power on to the master station.

16. PAC initialization of the I/O can now be accomplished.

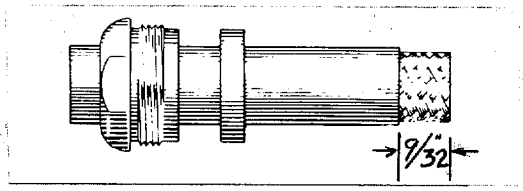
* The HCI is the interface between the HRMB board and the Process Automation Computer (PAC).

An exploded view of the detailed parts of the connector to be assembled is as follows:

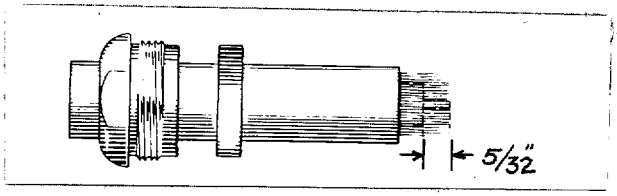


ASSEMBLY INSTRUCTIONS:

1. Place nut and gasket over the cable with the "V" groove toward the clamp, and cut off 9/16 inch of the jacket as follows:



2. Comb out braid and fold out. Cut off cable dielectric to 7/32 inch as shown below:



3. Pull braid wires forward and taper toward center conductor. Place clamp over braid and push back against cable jacket as follows:

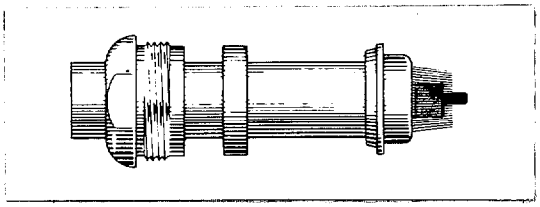
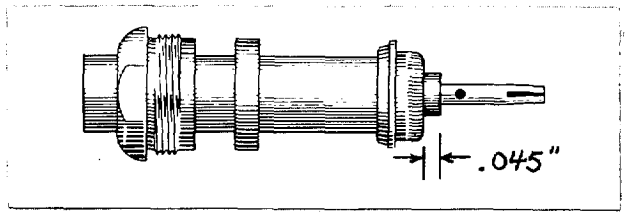


Figure 3-1
Attaching Connector to Customer Cable.

4. Insert cable and parts into connector body. Ensure that the sharp edge of clamp seats properly in gasket. Tighten nut.



NOTE

The dimension of spacing between the center pin and end-of-connector is essential to the proper functioning of the connector. Therefore, the tolerance shown is critical.

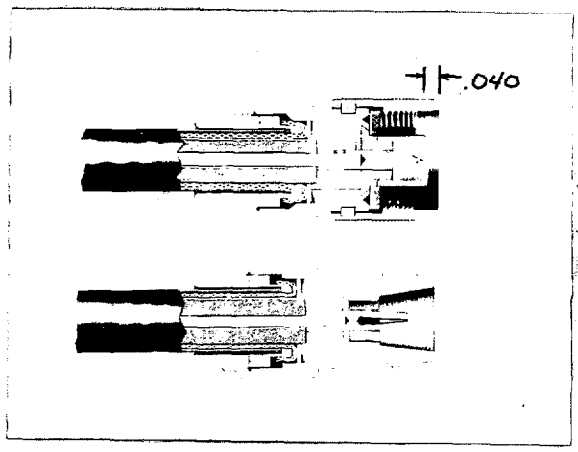


Figure 3-1
(Continued)

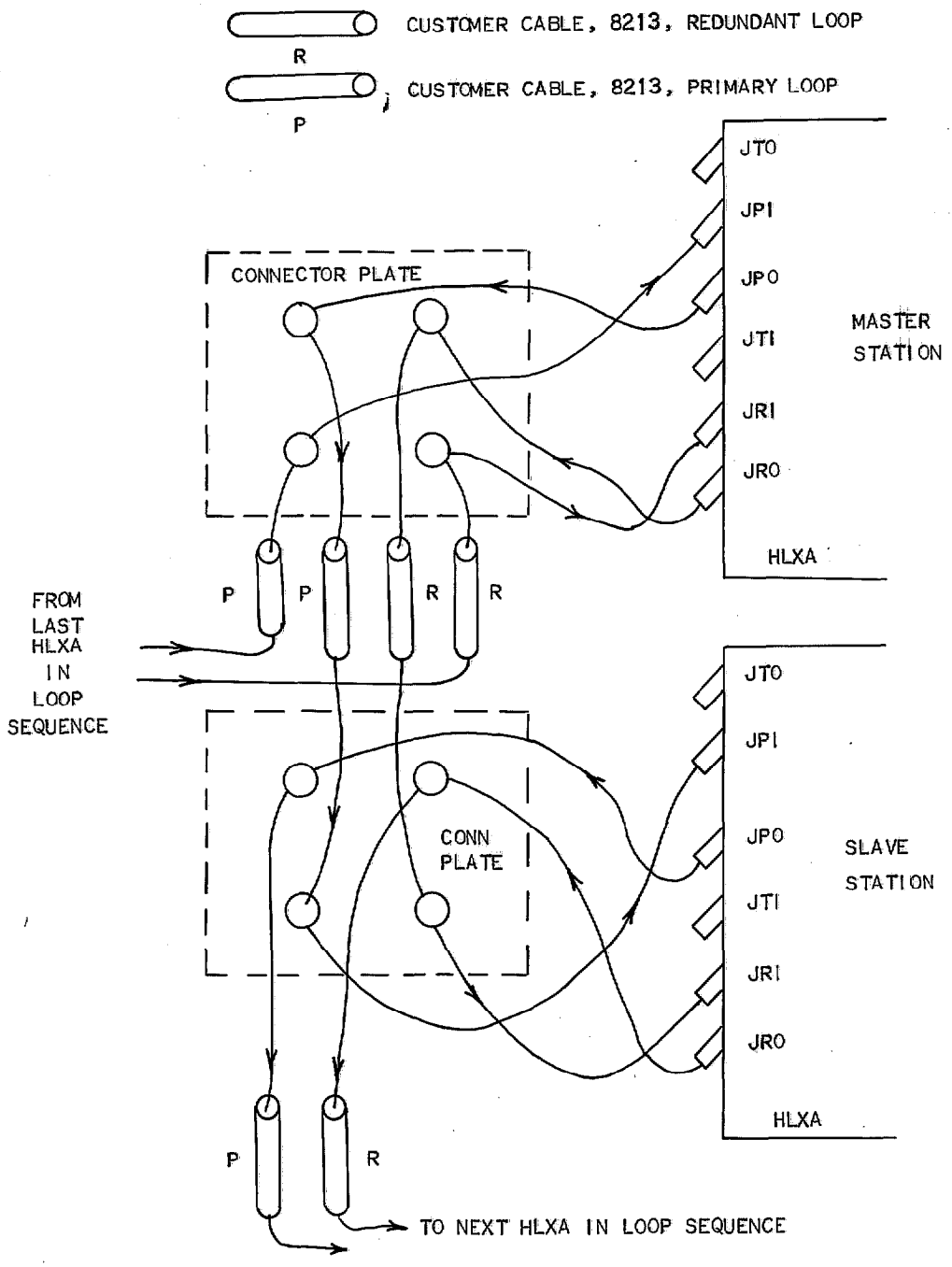


Figure 3-2.
Coaxial Cable Installation Diagram

POST INSTALLATION CHECKOUT PROCEDURE

The post-installation procedure verifies the capability to automatically switch over from primary to redundant cable. This procedure will also verify that the entire CSH system is operational.

1. Turn power off to master station and remove PAC interface card HCI*. Turn power on.

2. Disconnect the primary cable connection on the coax connector plate (See Figure 3-2).

3. Verify that the PCMC LED on the HHTB board is off. Also, verify that the LNK LED on HHRB board remains steadily lit.

4. Disconnect the redundant cable at the coax connector plate.

5. Verify that the PCMC LED on the HHTB board is presently blinking at a fast rate as the HMPC tries to establish viability, of one cable or the other, but finds neither cable to be acceptable.

6. Reconnect the primary cable at the coax connector plate.

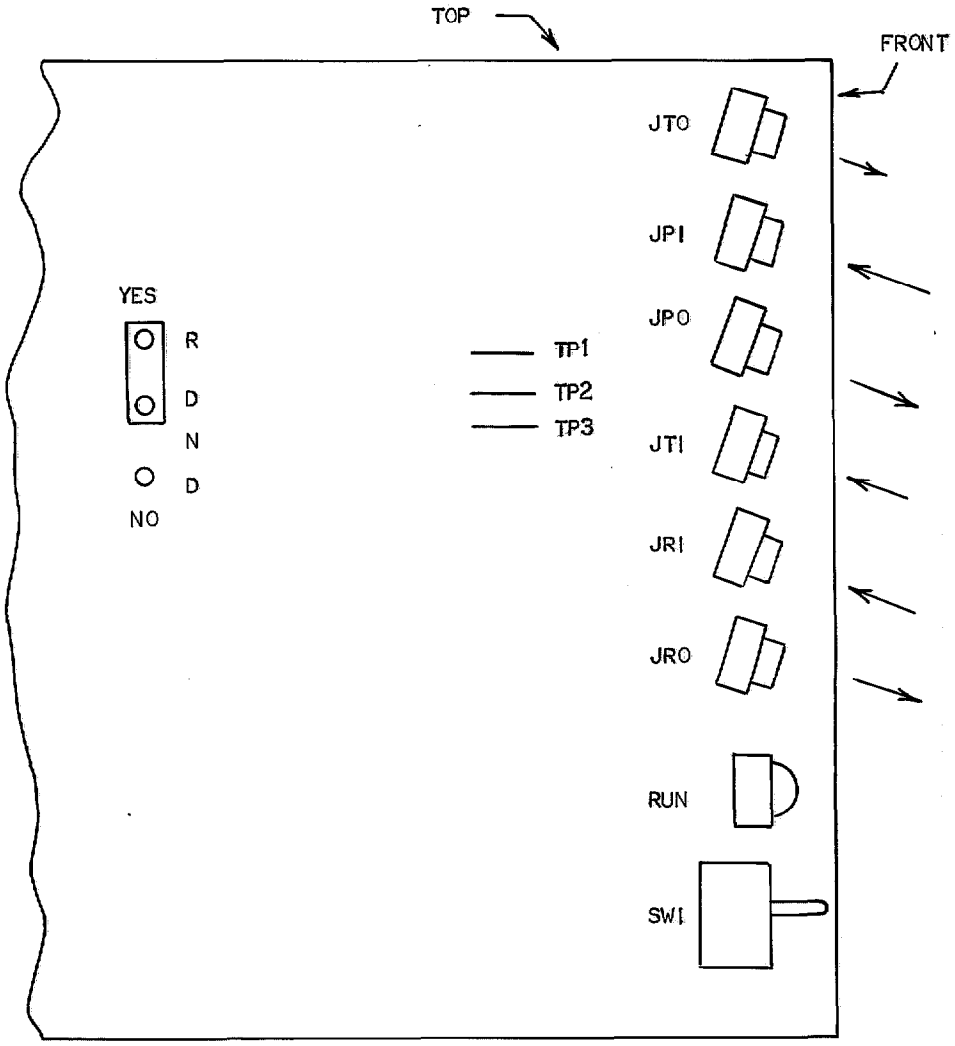
7. Verify that the PCMC LED is currently glowing steadily, then reconnect the redundant cable at the coax connector plate.

8. Repeat steps 2 through 7 (above) for each CSH loop in the system.

9. Randomly select known I/O input and output addresses and perform demand input (INPU) and output (OUTP) on them from the CRT terminal.

10. Turn power off to master station and insert PAC interface card HCI*. Turn power on.

11. Initialize the PAC. Normal process I/O should be cycling with XMIT LED on HHTB blinking.



ARROWS DEPICT DIRECTION OF MEASSAGE FLOW.
PRIMARY CABLE CONNECTS TO JPO, JPI.
REDUNDANT CABLE CONNECTS TO JRO, JRI.
HIGHWAY TEST MODULE CONNECTS TO JTO, JTI.

Figure 3-3
DS3800HLXA Installation Features

Table 3-1.
HLXA Indicators and Installation Variables

PWB: DS3800 HLXA

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumper RDNT	As shown in Fig.3-3	Slave will automatically switch to redundant cable when required. The slave will not switch when the jumper is in the NO position.
JTO, JTI	--	These are coax connectors for Highway Test Module.
JPO, JPI	--	These are coax connectors for primary cable connection.
JRO, JRI	--	These are coax connectors for redundant cable connection.
SW1	UP Position	Repeater is switched into CSH loop.
	DOWN Position	Coax bypasses repeater and connects repeater to connectors JTI and JTO.
RUN LED	Red	This is the indicator showing that the repeater is switched into the CSH loop.
TP1	Negative Pulse	Beginning of received message.
TP2	Sq Wave	First half of each received bit time.
TP3	DCOM	

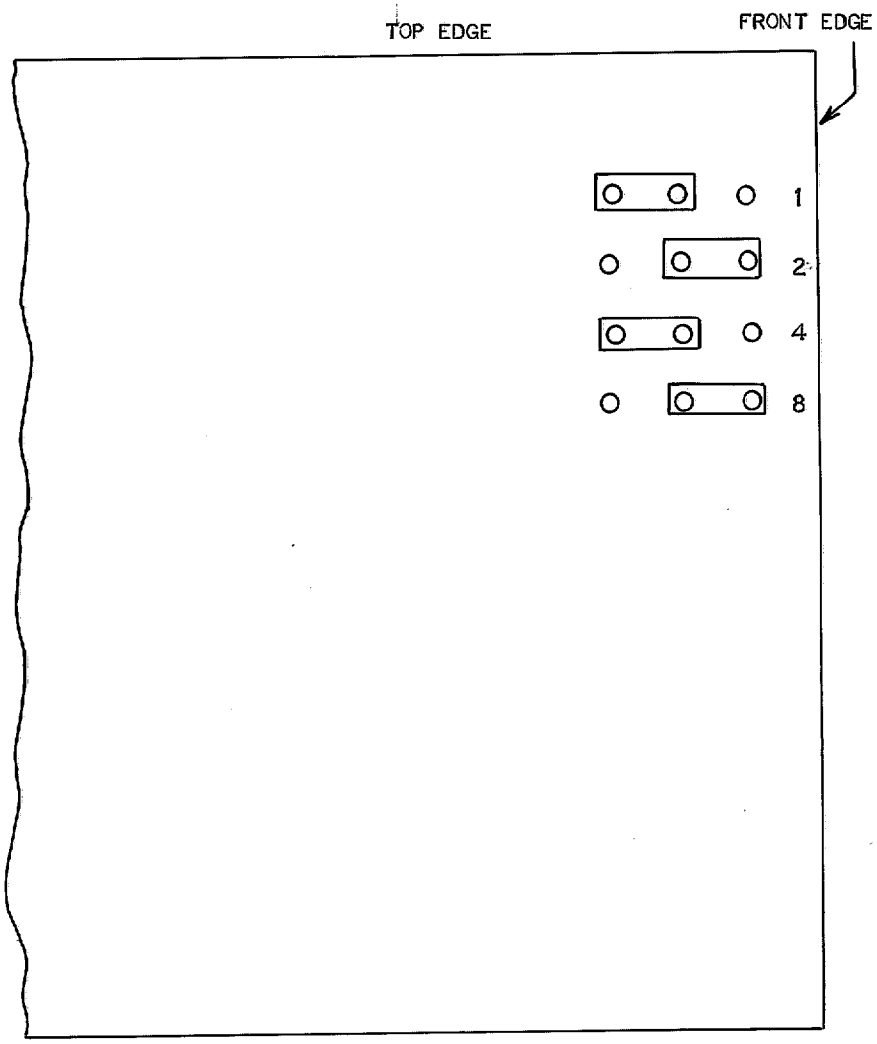


Figure 3-4
HLXB Installation Variables

Table 3-2.
HLXB Indicators and Installation Variables

PWB: DS3800HLXB

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumpers 1,2,4,8	As shown in Figure 3-4	Station jumpered to slave address "A" hex. Jumpers have no meaning in a CSH master station. The number imprinted next to the jumper's active position corresponds to the binary encoding of the station's slave address.

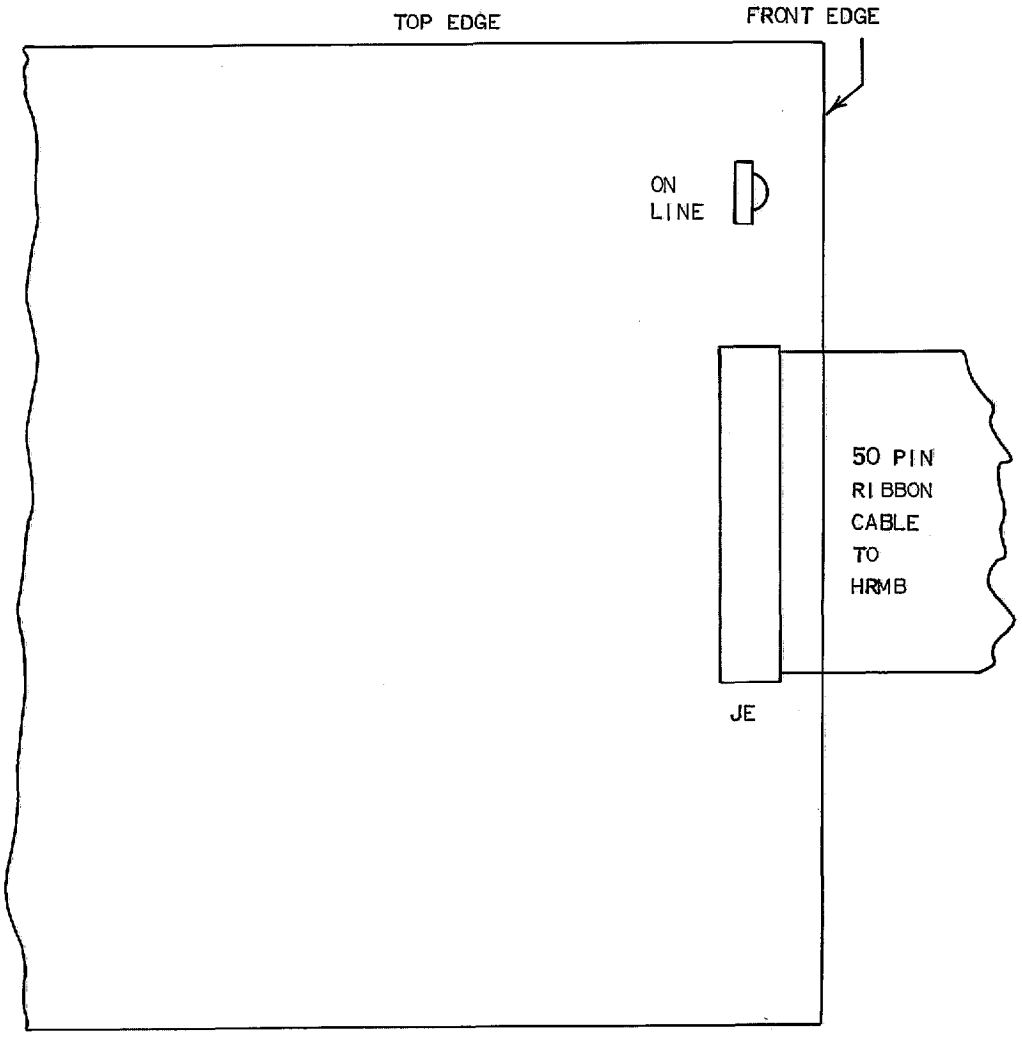


Figure 3-5
DS3800HCIA Installation Features

Table 3-3
HCIA Indicators and Installation Variables

PWB: DS3800HCIA

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
ON LINE LED	Amber	Process Computer (PAC) is not stalled.

****
* *
* CAUTION *
* *
****

DO NOT DEFEAT STALL DURING NORMAL CSH OPERATION, OTHERWISE, CERTAIN SAFETY FEATURES WILL BE DEFEATED, i.e., MODULATION CUTOFF DURING STALLOUT, OR WHEN THE MASTER STATION POWER SUPPLY FAILS.

CONN JE	--	50 pin connector to DS3800HRMB

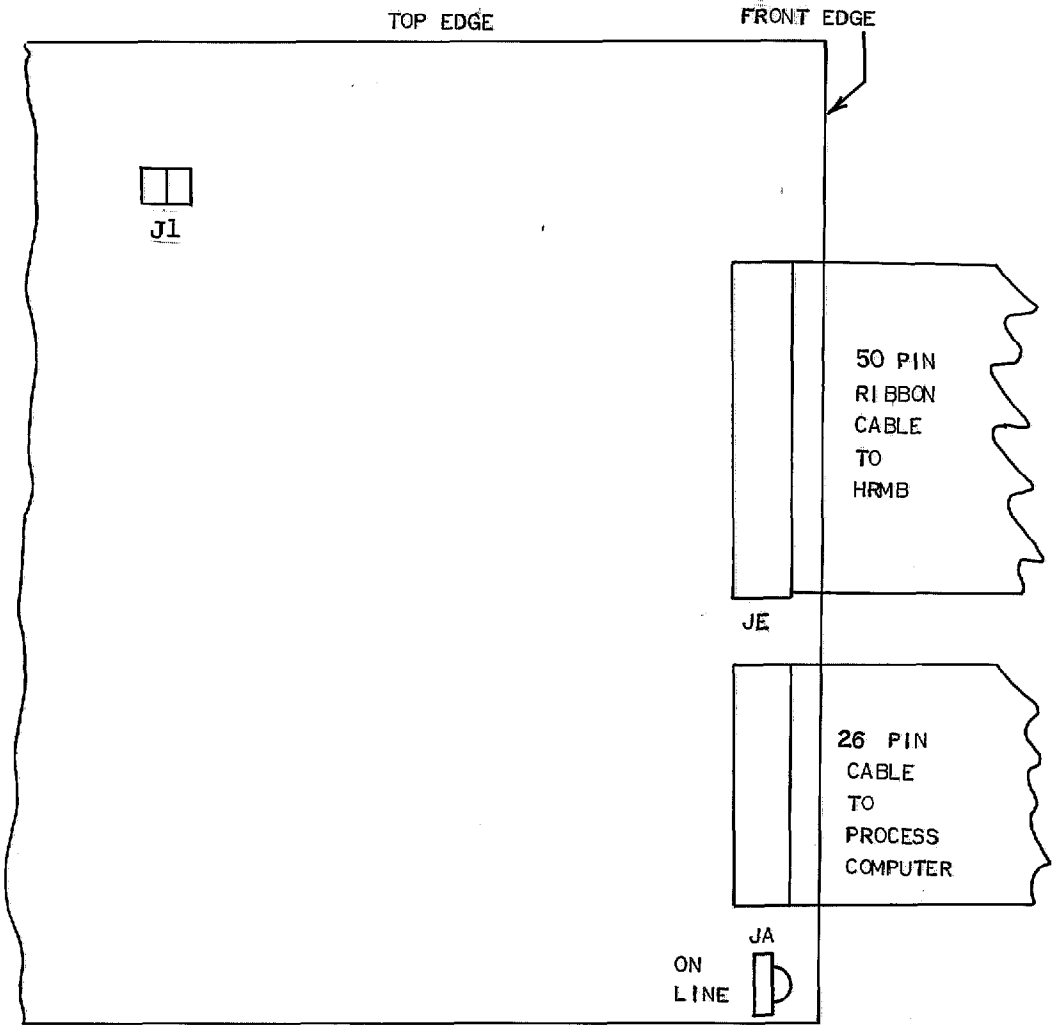


Figure 3-6
DS3800HCIB Installation Features

Table 3-4
HCIB Indicators and Installation Variables

PWB: DS3800HCIB

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
ON LINE LED	Amber	Process Computer (PAC) is not stalled.
		********** * * * CAUTION * * * ********** DO NOT DEFEAT STALL DURING NORMAL CSH OPERATION, OTHERWISE, CERTAIN SAFETY FEATURES WILL BE DEFEATED, i.e., MODULATION CUTOFF DURING STALLOUT, OR WHEN THE MASTER STATION POWER SUPPLY FAILS.
CONN JE	--	50 pin connector to DS3800HRMB
CONN JA	--	26 pin connector to PAC interface
J1	--	Used for card test. Must be installed for proper operation.

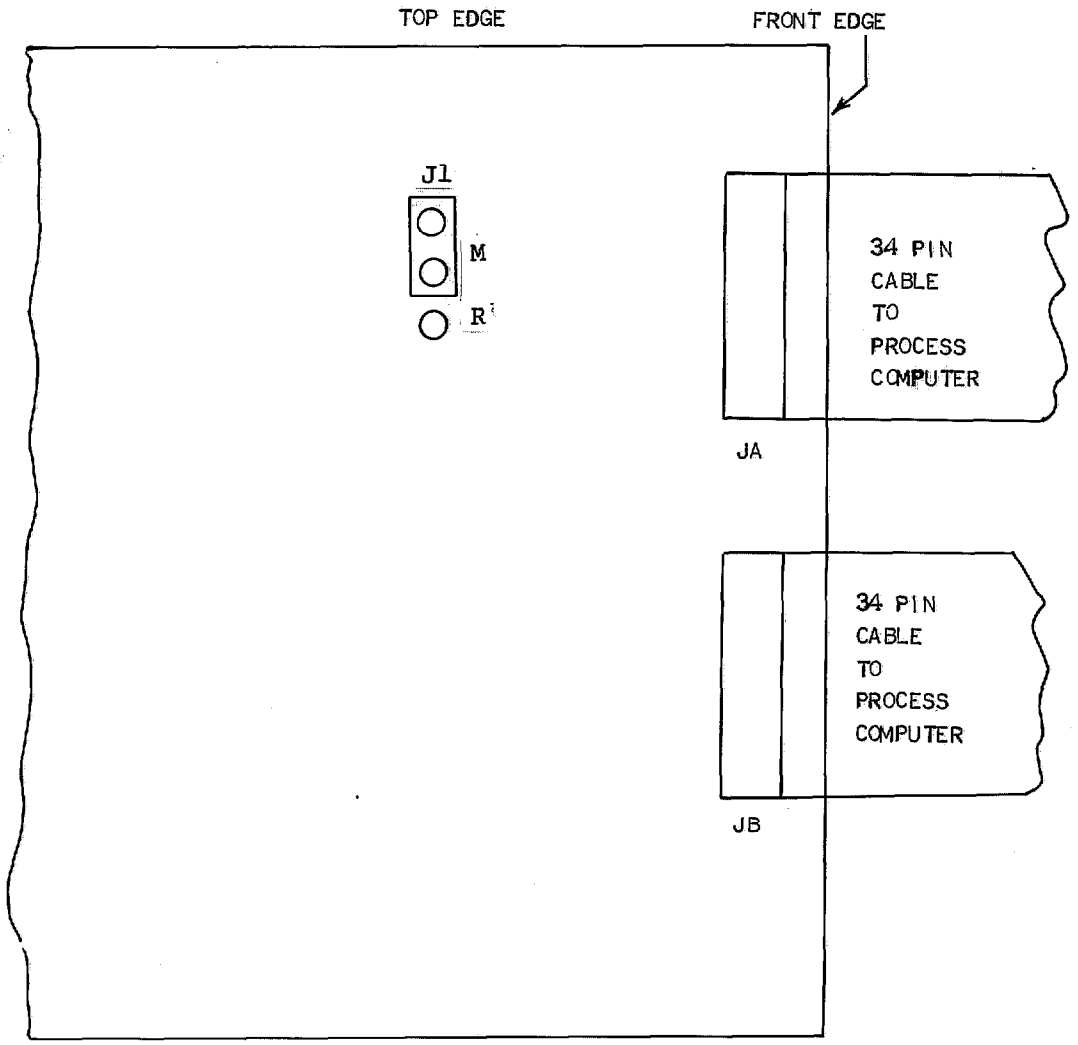


Figure 3-7
DS3800HDRA Installation Features

Table 3-5.
HDRA Indicators and Installation Variables

PWB: DS3800HDRA

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
CONN JA	--	34 pin connector to PAC interface (computer data input).
CONN JB	--	34 pin connector to PAC interface. (computer data output).
J1	R	R-Bus Interface
	M	Manual Non R-Bus Use

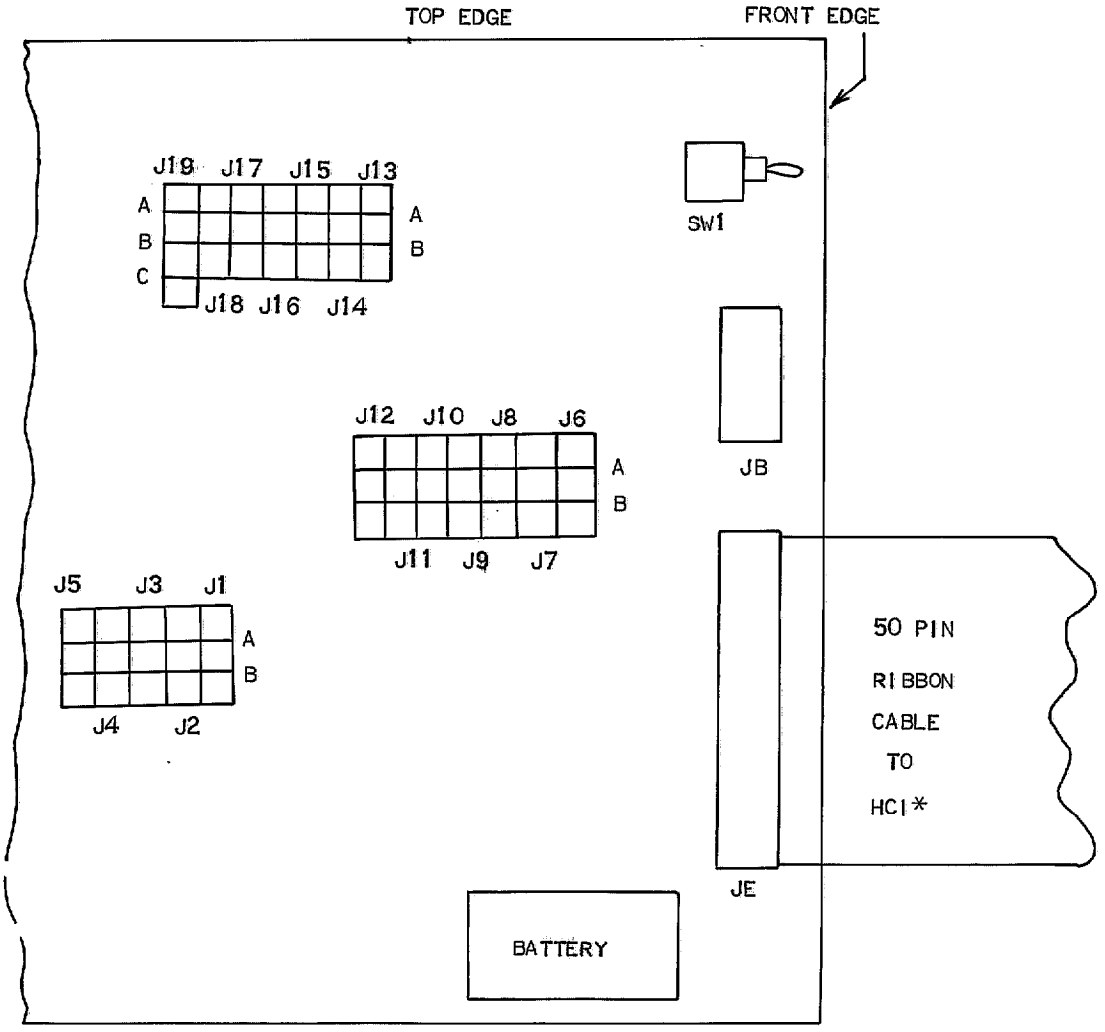


Figure 3-8
DS3800HRMB installation Features

Table 3-6
HRMB Indicators and Installation Variables

PWB: DS3800HRMB

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
CONN JE	--	50 pin connector to DS3800 HCI*
CONN JB	--	External battery / battery test.
SW1	UP/DOWN	Inserting or extracting board while power is on
SW1	Middle	Normal operation
J4-J1	A=1, B=0	L-BUS base address (X000H).
J5	A=0, B=F	L-BUS segment
J16-J18	AAA BBB	E and L-BUS select. L-BUS only.
J9-J6	A=1, B=0	E-BUS segment (X000H).
J12-J10	A=1, B=0	E-BUS base address (X000H).
J13-J15	AAA BBB	E and L-BUS select. E-BUS only.
J19	A B C	Internal clock test. Internal clock. External clock.

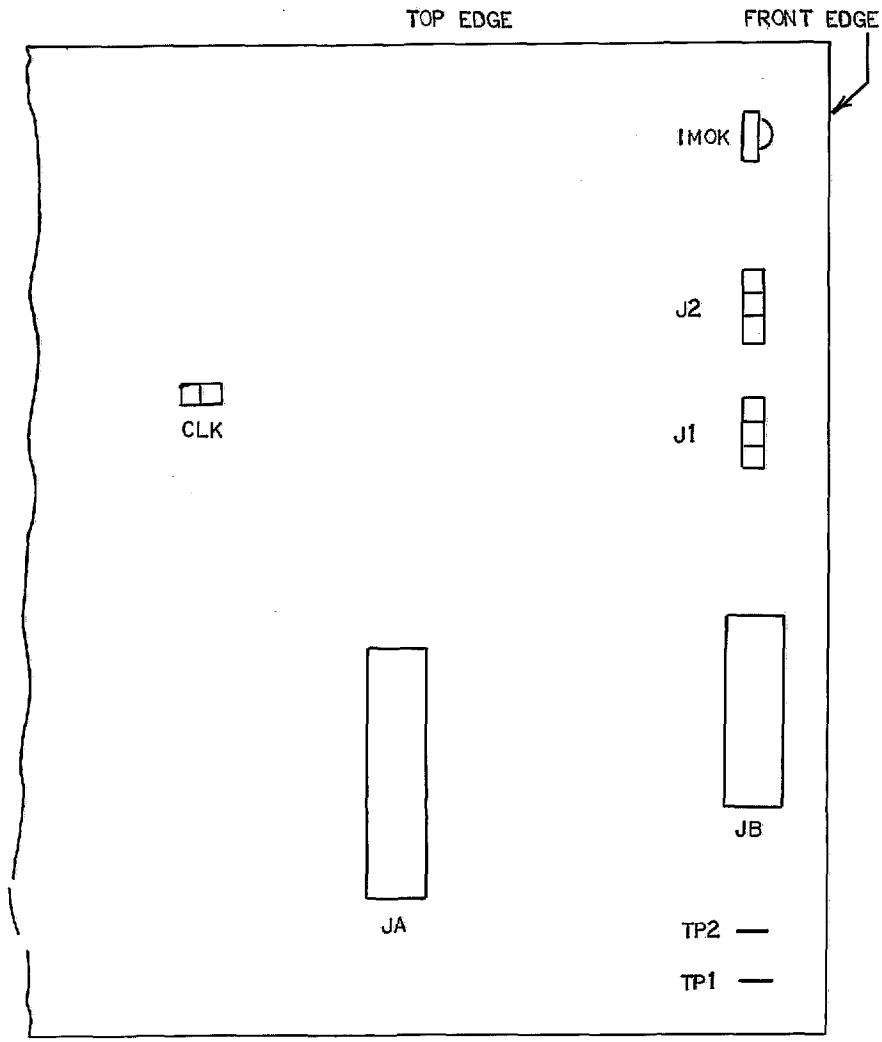


Figure 3-9
DS3800HMPC Installation Features

Table 3-7.
HMPC Indicators and Installation Variables

PWB: DS3800HMPC

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
IMOK LED	Amber	On when master station and PAC are cycling I/O.
J1	A	No connection.
	B	RS232 connector pin 7 (signal ground) to DCOM.
J2	TOP L	Locates the 16K bytes of on-board PROM at most significant address range of L-BUS.
	TOP E	Same as TOP L but for E-BUS.
TP1		Board reset.
TP2	DCOM	
CONN JB	--	RS232 connector.
CONN JA	--	Daughter board connector.
CLK jumper		Enables clock to RS232 interface (card test).

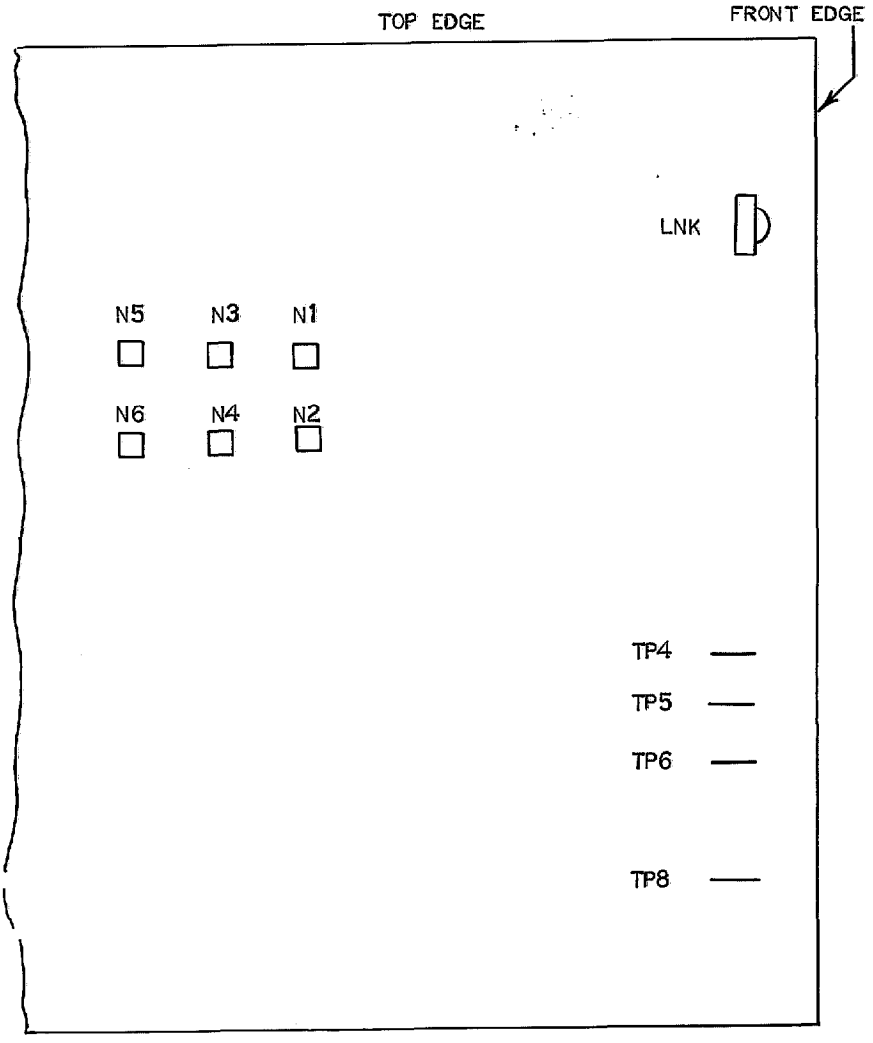


Figure 3-10
DS3800HHRB Installation Features

Table 3-8.
HHRB Indicators and Installation Variables

PWB: DS3800HHRB

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
LNK LED	Red	Lit when master station is able to modulate the complete CSH loop.
TP4	Sq. Wave	Returning message on CSH loop.
TP5	Pos. Pulse	Link error (HMPC Int 7).
TP6	Pos. Pulse	CSH message processing sync.
TP8	DCOM	
N1-N6		Used for card test only.

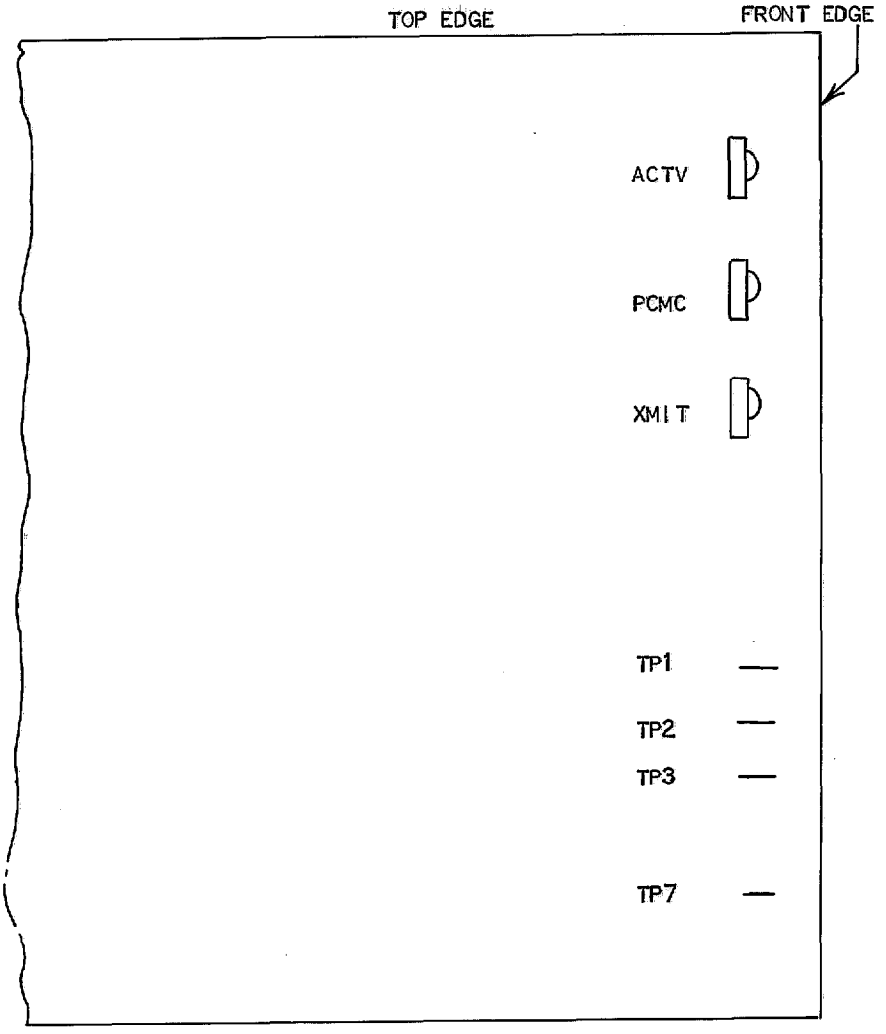


Figure 3-11
DS3800HHTB Installation Features

Table 3-9.
HHTB Indicators and Installation Variables

PWB: DS3800HHTB

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
ACTV LED	Red	Modulation turned on by HMPC.
PCMC LED	Red	Primary coax selected.
XMIT LED	Red	Non-idle message sent on CSH.
TP1	Sq. Wave	Outgoing message on CSH loop.
TP2	Pos. Pulse	Non-idle message transmitted on CSH.
TP3	Pos. Pulse	CSH message Sync Pulse.
TP7	DCOM	

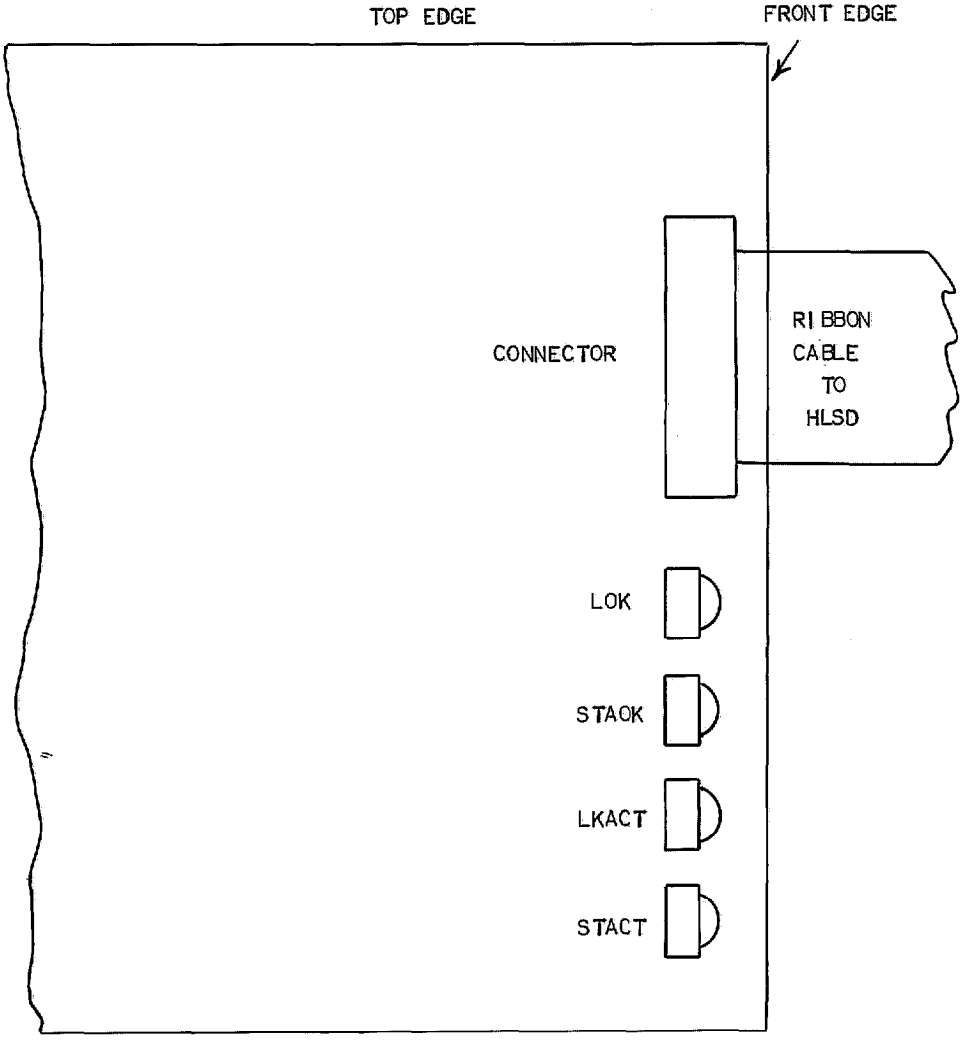


Figure 3-12
DS3800HLSC Installation Features

Table 3-10
HLSC Indicators and Installation Variables

PWB: DS3800HLSC

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
LOK LED	Amber	Comes on during power-up. Turns off when a faulty message is detected. It can only be reset by an "INQ" message once it has come on.
STAOK LED	Amber	Comes on during power-up. Turns off when an I/O device addressed does not respond or a parity error. It can only be reset by an "INQ" message once it has come on.
LKACT LED	Red	Legitimate modulation is passing through the CSH station.
STACT LED	Red	Messages addressed to this station are being received.
CONN	--	Connection to HLSD.

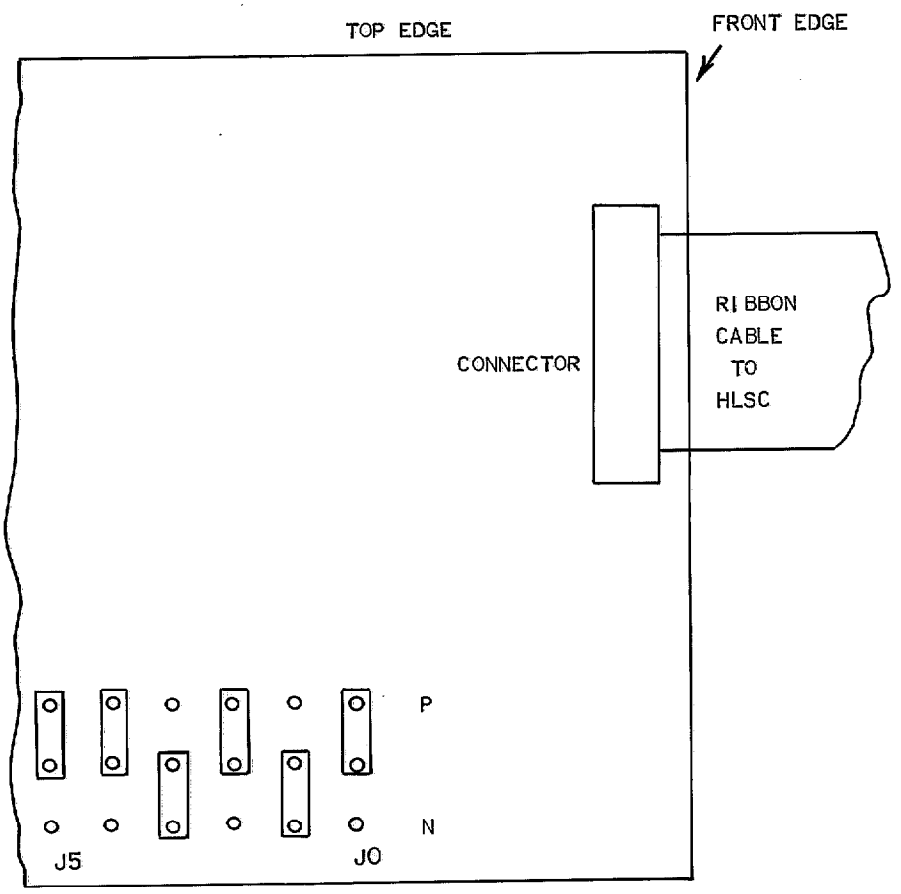


Figure 3-13
DS3800HLSD Installation Features

Table 3-11
HLSD Indicators and Installation Variables

PWB: DS3800HLSD

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Jumper JO	Shown on Figure 3-13	INTPT 0 (Bit 0) is selected for positive edge triggering.
Jumper J1	Shown on Figure 3-13	INTPT 1 (Bit 1) is selected for negative edge triggering.
Jumper J2	Shown on Figure 3-13	INTPT 2 (Bit 2) is selected for positive edge triggering.
Jumper J3	Shown on Figure 3-13	INTPT 3 (Bit 3) is selected for negative edge triggering.
Jumper J4	Shown on Figure 3-13	INTPT 4 (Bit 4) is selected for positive edge triggering.
Jumper J5	Shown on Figure 3-13	INTPT 5, 6, 7 (Bits 5, 6, and 7) are selected for positive edge triggering.

X X X X INQ	7 6 5 4 3 2 1 0	CRC

CSH MSG

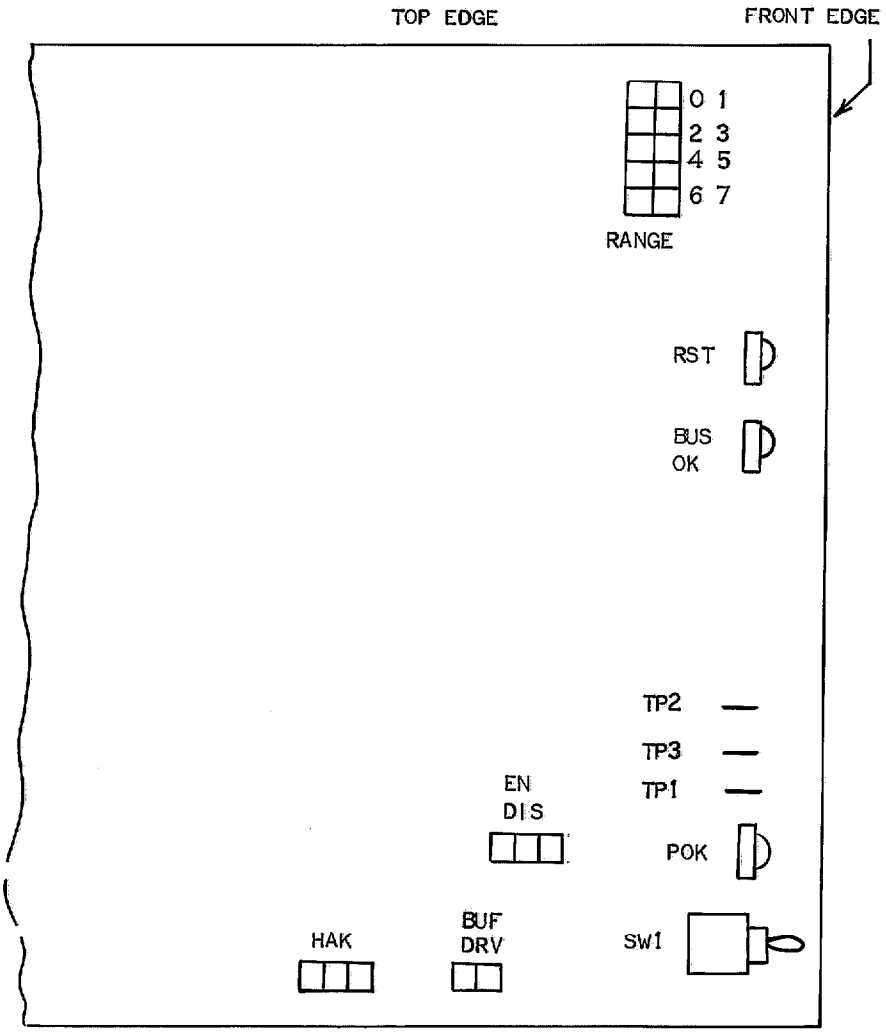


Figure 3-14
DS3800HPBC Installation Features

Table 3-12
HPBC Indicators and Installation Variables

PWB: DS3800 HPBC

ITEM	STATE	SIGNIFICANCE WITHIN CSH STATION
Range Jumpers		Used in buffer decode mode (stick) only. These jumpers select the upper hex digit of the I/O address range (32 addresses) selected.
RST LED	Red	I/O cards served by HPBC are being held reset.
BUS OK LED	Amber	I/O bus not shorted.
POK LED	Amber	Initially off. The LED latches on with the first good I/O transfer and will always reflect the status of the latest I/O transfer handled.
BUF DRV Jumper		Jumper for buffer drive mode (elect. module). Remove jumper for buffer decode mode (stick) and use as range jumper.
HAK Jumper	--	AUTO-Causes HOLD ACK to be automatically returned as handshake. It is commonly used in conjunction with a reduced bus. RHA-Another R-level controller will handshake HOLD ACK.
SW1	UP	Momentary I/O reset applied
SW1	DOWN	Forced I/O reset applied
SW1	MIDDLE	No I/O reset applied
TP1		Jumper to DCOM for R-level reset.
TP2	DCOM	
TP3		Logic "0" indicates HPBC has been addressed.
EN/DIS jumper		Enable/disable operation of POK LED.

--SECTION 4--

CSH MASTER STATION SOFTWARE USER'S GUIDE

This user's guide describes the firmware which resides in the Control Signal Highway (CSH) master station controller. This master station is designed to interface with any type of process automation computer (PAC) via a dual ported random access memory card (HRMB), which is included in the master station module. Additionally, an interface card (HCI) specifically designed to interface with the particular PAC being used is required. A complete description of the master station hardware is available in Section 2 of this document. It is assumed that the reader is familiar with the basic operation of the CSH loop and the hardware required.

Through this PAC-master station interface, the PAC can issue commands to control the CSH loop. It is the function of the firmware described here to interpret these commands and operate the Control Signal Highway. A CRT terminal connection to the master station is also available to perform these functions.

Master Station Firmware Structure and Operation

This software system is based on a multi-tasking operating system (MTOS) acquired from Industrial Programming Inc., Greenvale, New York. The operating system is a 'stripped down' version in order to save memory space and because only the basic operating system capabilities are needed.

The CSH Controller firmware consists of approximately 14K bytes of programs. Approximately 5K is consumed by the operating system and terminal driver (MTOS), which is assembly language code. The remaining 9K is PLM-86 except for the CSH I/O drivers which are written in assembly language for execution speed reasons.

Table 4-1
HRMB Data Structure

The HRMB RAM memory card is configured as shown below. The remainder of this document will refer to this memory area as: HRMB Offset (word).

HRMB Offset (hex)		
0	function code	contains code of function being requested by the PAC or terminal
1	station number	station number being addressed
2	page/group	page and group within above station
3	interval	intervals to be processed during current sweep (sync mode only)
4	spare	
5	stations test I/O	stations to be tested during test function
6	system parameter pointer	pointer to system parameters
7	control word pointer	pointer to I/O control words
8	output data pointer	pointer to output data words
9	input data pointer	pointer to input data words
A	spare	
B	spare	
C	system status flags	see table 4-2
D	station input disabled	a corresponding bit for each station is set to indicate station input disabled
E	station output disabled	same as above for output
F	system error flags	see table 4-3
10	station CRC errors	corresponding bit is set for a station which has a CRC error
11	station unhealthy	corresponding bit is set for a station which has a station error
12	station link error	corresponding bit is set for a station which has a link error
13	station error counters	these counters count any errors which occur for each station (1-F)
23	station interrupts	these words record any station interrupts received by a station (1-F)
33	demand input value	word received as a result of a demand input function
34	demand output value	word transmitted by demand output function
35-FFF	Unassigned	configured by PAC

Table 4-2
System Status Flags

HRMB Offset $0C_H$ Bits 0-15

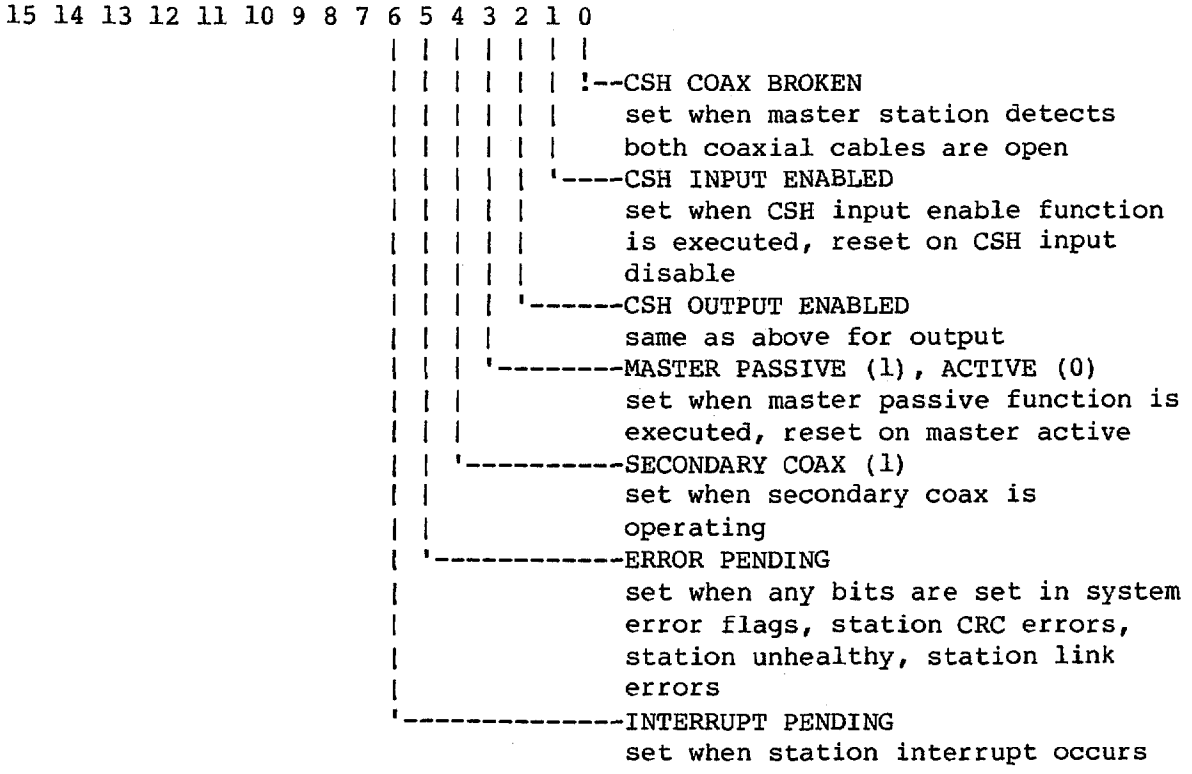
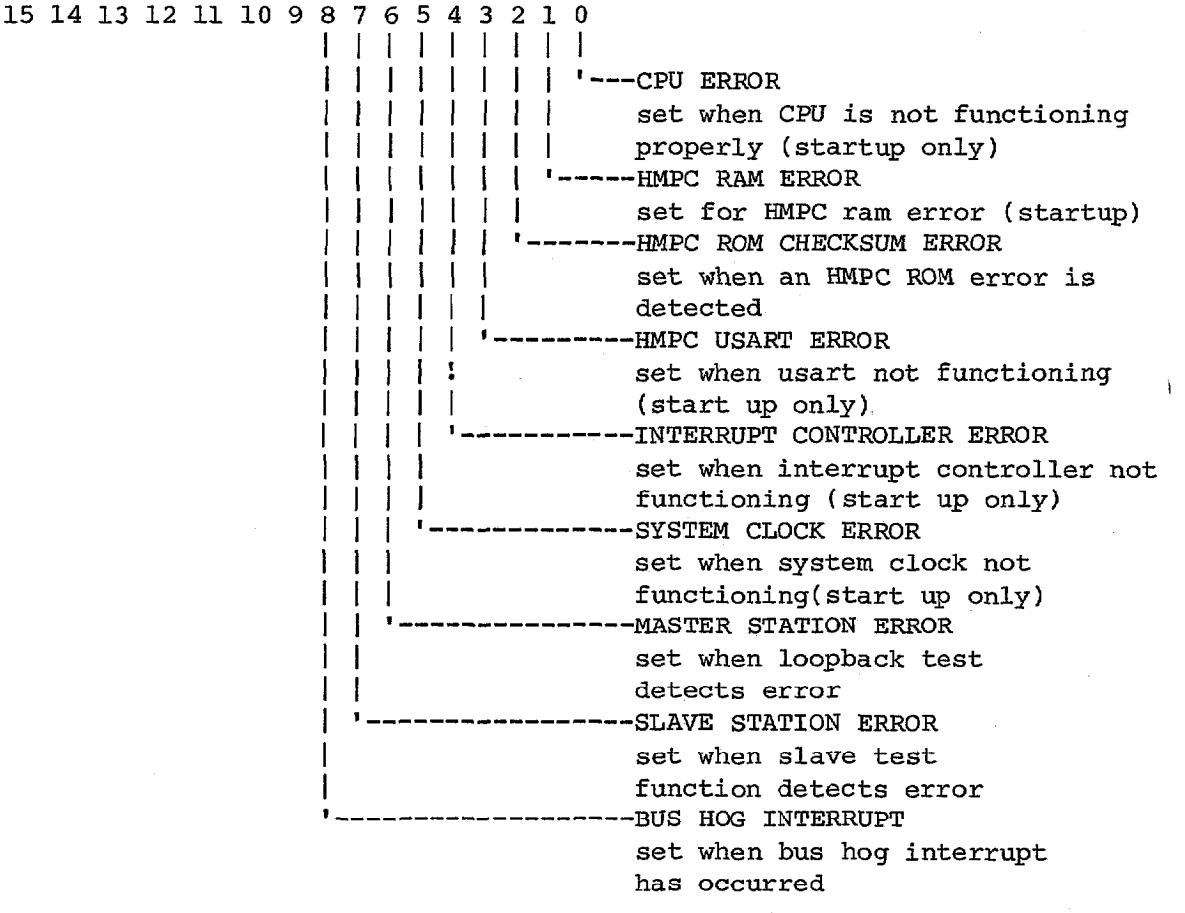


Table 4-3
System Error Flags

HRMB Offset $0F_H$ Bits 0-15



CSH System Parameters

The system parameters are written into the HRMB by the PAC before the build system parameters function can be executed. They are as follows:

HRMB Offset 06_H = pointer to parameters

0	SYNC/ASYNC (1)	1 indicates synchronous mode being used
1	not used	spare
2	number of intervals(2)	maximum of 8 intervals in system(0-7)
3	interval start(3)	number of sweeps to occur before initial execution of control words assigned to each interval (8 intervals, 1 word per interval, minimal value = 1)
B	interval refresh(3)	number of sweeps to occur between execution of control words assigned to each interval (8 words, 1 per interval)

(1) synchronous means that all functions, including I/O are being done individually as requested by the PAC.

asynchronous means that I/O is being done by the master station automatically based on the intervals and control tables supplied by the PAC.

(2) An interval is a group of control tables. Certain control tables may be assigned to execute in an interval. If more than one interval is scheduled to run during the same sweep, the lower numbered interval has priority. If zero, no I/O will be executed.

(3) Example: Four intervals are desired (0,1,2,3) with an interval sweep map 01-02-03 to be run asynchronous. The HRMB word offset for the system parameter table would be:

```
WORD  0  =  0 (async)
      2  =  4 (4 intervals)
      3  =  1 (interval 0)
      4  =  1 (   "     1)
      5  =  2 (   "     2)
      6  =  3 (   "     3)
      B  =  1 (   "     0)
      C  =  3 (   "     1)
      D  =  3 (   "     2)
      E  =  3 (   "     3)
```

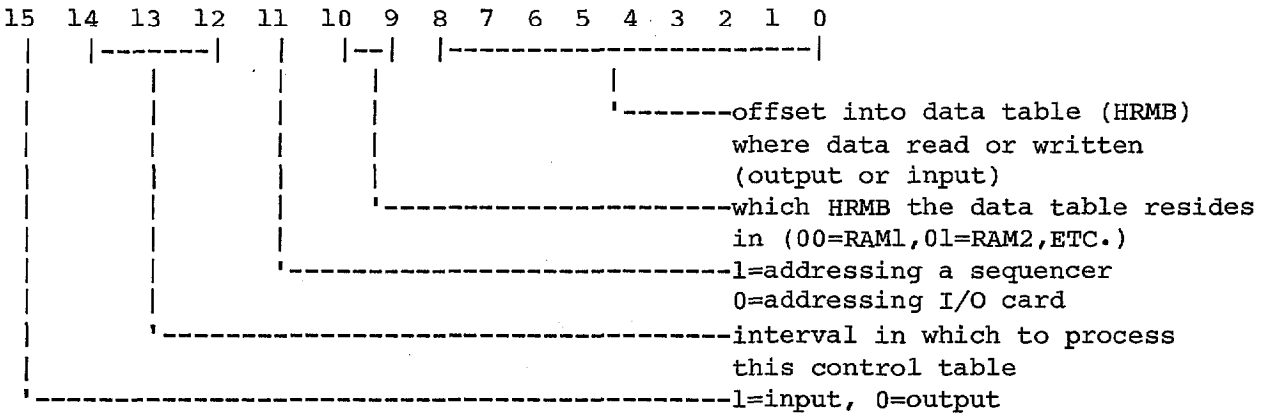
CSH Control Tables

The control words must be written into the HRMB before the build control word function can be executed. They must be grouped by interval in ascending interval order. Each control word is made up of 3 words. A word of FF_H indicates the end of the table. They are configured as follows:

byte offset (word 1 1,0/word 2 3,2/word 3 5,4)

0	number of sequential groups (I/O addresses) to be addressed *
1	slave station to be addressed (01-0FH) - bits 4-7 / bits 0-3 is slave control table number (PAC use only)
2	starting group to be addressed in sequence (LS address byte)
3	Page being addressed (MS address byte)
5-4	Offset into input or output data table + flags (see below)

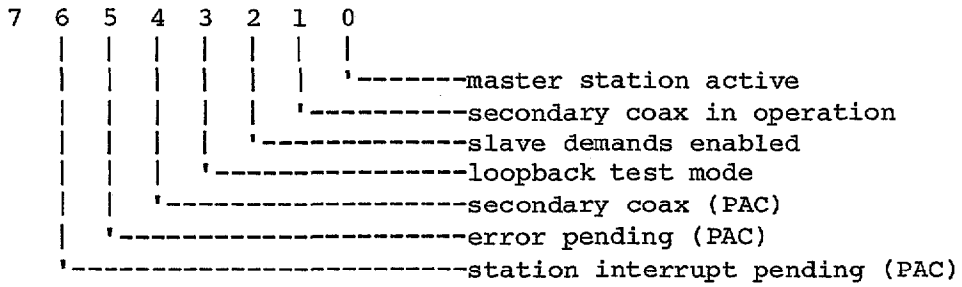
data table offset (control table word 3)



* If zero, specific control table will be bypassed.

HMPC Direct Output Bits

There are 8 direct data bits output to HMPC I/O port 0BFE2H, which control the master station hardware and are also available to the PAC:



Power-up Sequence and Initialization

Power-up, or reinitialization, starts in an MTOS module, INIT. The first HRMB(00) is cleared and the first location (adr 0) is set to 0FFFFH, to indicate that initialization is taking place. All hardware initialization and MTOS initialization is done here based on the parameters defined in another MTOS module, XDATA. In addition the following diagnostics are performed at initialization:

1. CPU check (1 blink)
2. HMPC RAM check (2 blinks)
3. HMPC ROM Checksum (3 blinks)
4. Peripheral interrupt controller check (4 blinks)
5. Clock check (4 blinks)
6. Bus hog interrupt check (4 blinks)

If a failure occurs on any of these checks, the "IMOK" LED on the HMPC board will blink the specified number of times, pause, and repeat this blinking sequence continually. The system is inoperable at this point. In addition, the appropriate flags are set in HRMB system error flags word, which is available to the PAC.

If all initialization and diagnostics complete successfully, a start-up task (STARTUP) will be started. This task programs the CSH link cards, enables the system terminal and schedules a low priority task which will continually check the HMPC ROM checksum.

At this point, the first location of the HRMB is cleared, and the system is waiting for commands via interrupts from the system terminal or PAC.

Process Automation Computer Interface

The PAC interfaces to the master station through an interface board (HCI*) and a dual ported RAM board (HRMB). The only means for the PAC to control the master station are: 1) The PAC can reset the master station; this is the same as a power-up situation. 2) The PAC can interrupt the master station processor (HMPC). 3) The PAC may (depending on its capability) read 3 digital output bits from the master station. 4) Data can be passed through the dual ported RAM (HRMB).

Therefore, the normal interface is this: The PAC places a function code and parameters in the HRMB and interrupts the HMPC. The master station performs the desired function and places any returned data or error codes in the HRMB. When the function is complete, the function code in the HRMB is cleared to 0 by the HMPC, indicating to the PAC that the function has been completed. The PAC may now read any return data in the HRMB and may read the HMPC direct output bits.

Terminal Interface

The RS-232 port on the HMPC will accept (but does not require) a standard RS-232 terminal operating at 1200 baud, odd parity. All functions available from the PAC are available when the PAC is doing operations or has specified asynchronous on-line I/O to be done.

All functions and required parameters are listed in Table 4-4. The parameters must be specified in hexadecimal format, in the order in which they are required. For example, to do an input from station F, page 0, group 42, the terminal operator would enter:

```
INPU,F,0,42 (Carriage return)
```

These parameters are placed in the HRMB, just as the PAC does and the same task is scheduled to process the function.

After the function is complete, the results are displayed as shown in Figure 4-1.

Table 4-4
CSH FUNCTION CODES

FUNCTION	MNEMONIC	CODE	REQUIRED PARAMETERS
no action	none	0	none
build system param(#)	SYSP	1	none
build control tbls(#)	BCTL	2	none
normal I/O	NORM	3	interval
slave test	TEST	4	data out, stations**
demand input	INPU	5	station, page, group, c
demand output	OUTP	6	station, page, group, data, c
station in disable	SIDI	7	station
station out disable	SODI	8	station
station in enable	SIEN	9	station
station out enable	SOEN	10	station
group in disable	GIDI	11	station, page, group
group out disable	GODI	12	station, page, group
group in enable	GIEN	13	station, page, group
group out enable	GOEN	14	station, page, group
CSH input disable (#)	CIDI	15	none
CSH input enable	CIEN	16	none
CSH output disable (#)	CODI	17	none
CSH output enable	COEN	18	none
active master	AMAS	19	none
passive master (#)	PMAS	20	none
primary coax	PCOA	21	none
alternate coax	ACOA	22	none
no action	none	23	none
master loopback (#)	LOOP	24	data, c
clear flags	CLEF	25	none
clear system errors	CLEA	26	none
verify HRMB (#)	VERI	27	none
memory display	MEMD	28	address*, # of words (HEX)
memory write	MEMW	29	address*, value

\# do not use with process control on line
c if continuous mode is desired #
* address = offset, segment (byte address)
segment = 0
offset = 3000H HRMB 00
5000H HRMB 01
7000H HRMB 02
9000H HRMB 03
** Refer to description of slave test

DATA	SYSTEM	STATION	STATION ERROR COUNTERS
IN XXXX	STATUS XXXX	INDIS XXXX UNHLTY XXXX	1- XXXX XXXX XXXX XXXX XXXX
OUT XXXX	ERRORS XXXX	OUTDIS XXXX LINK XXXX	6- XXXX XXXX XXXX XXXX XXXX
		CRC XXXX	B- XXXX XXXX XXXX XXXX XXXX

DATA IN/OUT - displays HRMB input value / output value for Demand I/O
SYSTEM STATUS - displays hex value of HRMB system status errors
SYSTEM ERRORS - displays hex value of HRMB system errors
STATION INDIS/OUTDIS - displays hex value of HRMB CSH stations disabled
STATION CRC/UNHLTY/LINK - displays hex value of HRMB crc/unhealthy/link errors
STATION ERROR COUNTERS - displays HRMB error counters (1-5,6-A,B-F)

Figure 4-1
CRT Function Display

CSH Function Codes

All of the following functions are available to either the PAC or terminal except for the memory display and write which are only available to the terminal. Table 4-4 illustrates for each function code, the mneumonic to be entered on the terminal and the required parameters.

Build System Parameters

This function simply moves the initial and refresh interval values from the HRMB memory. The system parameters must be in the system parameter table before requesting this function.

Build Control Tables

This function reads the control tables written into the HRMB by the PAC and manipulates them into a format more easily used by the master station. If this function is to be invoked from the terminal, the tables must be written into the HRMB using the memory write function.

The tables are built in HRMB memory as follows:

1) Input and output words are grouped into the sequence desired for process control. Normally input and output tables are not interleaved.

2) Within the tables, the words are grouped by interval, starting with interval 0. A table of indexes, one index for each interval, is kept for each control word table (one for input, one for output). Each interval index points to the first control word to be executed in that interval.

NOTE: The speed of the CSH I/O is dependent on the number of control tables required and the number of I/O addresses. The approximate time it takes to process one interval sweep can be determined as follows:

```
# of Control Tables X 230 usec +
# of I/O addresses in Interval X 50 usec +
600 usec (system overhead per interval sweep) =
Time per Interval Sweep
```

The exact interval sweep time can be measured in the following manner:

1. Connect an oscilloscope to DS3800HMPC, Pin 80 (1D02).
2. Verify that no errors exist on the CSH loop by use of the diagnostic CRT.
3. The time between pulses equals the interval sweep time required for each interval specified by system parameters.

The control words built in the HMPC RAM memory now look like this:

byte offset

0	flags see below	
1	number of groups	number of sequential addresses to be addressed this burst
2	page page to be addressed	
3	page-station command	CSH command to send page address
4	group group to be addressed	
5	group-station command	CSH command to send group address
6-7	buffer pointer	pointer to data area in HRMB where data is stored or retrieved

flags:

```
7 6 5 4 3 2 1 0
| |
| '-------------I/O card (0)/sequencer (1)
'---------------enabled (0)/disabled (1)
```

Normal I/O

This function allows for the one-time execution of all control words within the interval specified by the interval HRMB word (synchronous I/O only). This is valid only after system parameters have been initialized and control tables are built.

Demand I/O

These functions perform input or output to the address specified. The results of these functions are displayed in the 'data' fields of the display, if initiated from the terminal. The result is kept in HRMB demand input or output value words.

Station Enable/Disable

This function enables or disables inputs or outputs to the station specified. The corresponding bit in the HRMB station input or output disable is set if disabled, reset if enabled. This function is accomplished by setting or resetting the disabled bit in any control words which address the station.

Group Enable/Disable

This function enables or disables input or output to a particular group with the station and page specified. This is done by setting or resetting the disabled bit in any control words which address the group. This function effects all addresses in all control word tables containing the group selected.

CSH Enable/Disable

This is done by setting (enabled) or resetting (disabled) the appropriate bit in HRMB system status flags. If CSH input or output is disabled, no asynchronous input or output is done. However, I/O can still be done from the terminal or synchronously from the PAC.

Master Station Active/Passive

If the master station is made passive, modulation on the link is repeated by the master station. The station is made passive by resetting direct output bit 0 to HMPC I/O port 0BFE2H. Bit 3 in HRMB system status flags is set when passive, reset when active. This allows two master stations to be on the same CSH loop at once with loop mastership being software controlled.

Switch Coax

This function switches operation to the coax specified. It is accomplished by setting (alternate) or resetting direct output bit 1 to port 0BFE2H. Also, bit 4 in HRMB system status flags indicates the current operating coax.

Clear Errors

This function clears the following words in the HRMB:
1) error counters
2) station CRC errors
3) station unhealthy errors
4) station link errors
5) system error flags
6) station interrupts
7) interrupt pending and error pending bits in system status flags
8) interrupt pending and error pending direct data bits to port 0BFE2H

This function also flags the HMPC that PAC diagnostics are complete at initialize time to allow the HMPC to write to the HRMB.

Clear Error Flags

This function clears bits 0, 5, 6 in the system status flags.

HRMB Verify

This test verifies the PAC-HRMB-master station interface. Before requesting this function, the SPAC should write a pattern throughout the HRMB to be checked. Also, a code must be supplied in the second word of the primary HRMB to indicate which HRMB is to be checked (0-3). Each word of the HRMB is checked that each word is identical, then complements each word. If an error is detected, the second location of the primary HRMB is set to all ones. The PAC should now check that each location has been complemented.

Memory Display

This function displays words of memory starting from the byte address specified. The address must be specified in standard 8086 format:

offset, segment, # of words (FFH MAX)

Memory Write

Memory write writes the one word of memory specified to the byte address specified. The format for the parameters is:

offset, segment, data

Asynchronous I/O

Following the build control word function, the asynchronous I/O scheduler is scheduled to run. From this time on, the I/O scheduler runs continuously, repeating the following steps:

1) Each interval counter is decremented. For all interval counters that go to 0, a corresponding bit is set in the HRMB I/O intervals and the counter is refreshed with the interval refresh value specified in the system parameters. These intervals are now scheduled to run in this sweep.

2) If the loop is in sync, I/O is done for all control blocks assigned to any intervals whose I/O interval flags were set as a result of their counters decrementing to 0. These inputs and outputs are done only if the CSH input and/or output is enabled.

3) Stations interrupts are enabled for a short period following I/O to allow them to be serviced.

NOTE: Each slave station can remember the last page addressed within that station. Therefore, a page address is sent to a station, only if it is different from the previous page addressed.

Interrupt Processing

There are four interrupts in this system which are directly connected with the master station operation. They are as follows:

1. PAC Interrupt - Interrupt 4, Vector 30

   This interrupt is generated by the PAC through the HCI* card. When the PAC needs the master station to perform a function, it should write a function code and required parameters into the HRMB and then interrupt the master station.

Upon acknowledging this interrupt, the master station will schedule the function processor task to process the requested function. Upon completion of the task, the function code will be reset to 0. This interrupt is enabled at all times except when active data messages are being transmitted on the link.

Broken Coax - Interrupt 5, Vector 37

2. This interrupt is generated by the master station when a broken coax cable is detected by the master station. A task is scheduled which determines which cable the system was operating on (primary or redundant) and then directs the master station to switch to the other cable. This is done by setting a direct output bit to I/O port 0BFE2H. The status of which cable is being used is kept in the HRMB system status flags. Additionally, direct output bit 4 is set or reset and sent to port 0BFE2H to indicate, to the PAC, which cable is being used.

   This interrupt is enabled at all times except when active data messages are being transmitted on the loop. If for any reason, both coaxial cables remain open, the controller will indicate to the PAC an open coax condition.

3. Slave Demand - Interrupt 6, Vector 38

   This interrupt is generated by the master station when it detects that one or more slave stations on the loop is requesting attention. The slave may request attention for any of three reasons:

1) The slave detects a CRC error in a non-idle message. This is termed a'link error'.

2) The slave does not receive a response, or a parity error is detected, from an I/O device being addressed by the CSH. This is termed a 'station unhealthy'.

3) A special service request is made by an I/O device being served by the slave station.

   When this interrupt occurs, an idle message is read from the CSH which indicates which slave stations are requesting service. The master station sends an inquiry message to each demanding slave, one at a time, each one returning a status word. The master then ackowledges the demanding station and moves on to the next demanding station.

If the status word from the slave indicates that one of the two errors has occurred: 1) A bit corresponding to the station is set in HRMB station unhealthy or HRMB station link error, depending on the error. 2) The error counter in the HRMB for that station is incremented. 3) The error pending bit in HRMB system status flags is set. 4) The error pending direct data bit is set and output to port 0BFE2H.

If the status word from the slave indicates that a remote interrupt has occurred: 1) The status word is placed in the corresponding HRMB station interrupts. 2) The interrupt pending bit in HRMB system status flags is set. 3) the station interrupt direct data bit is set and output to port 0BFE2H.

This interrupt is only enabled for a short time at the end of each sweep.

4. CRC Interrupt - Interrupt 7, Vector 39

The master station generates this interrupt when it detects a CRC error in a message returning from a slave. It is enabled only when active data messages are being transmitted on the link.

When the processor acknowledges this interrupt, the sequence of messages currently being sent is stopped, restarted and the HRMB error counter for that particular station is incremented. If the interrupt occurs on the second attempt, the processor moves on to the next sequence to be transmitted. Additionally: 1) The HRMB station error counter is incremented again. 2) The corresponding bit in HRMB CRC errors is set. 3) The error pending bit in HRMB system status flags is set. 4) The error pending direct data bit is set and output to port 0BFE2H.

If this interrupt occurs while sending a sync message to a slave, it is assumed that the station cannot be synchronized.

If this interrupt occurs while processing a slave demand interrupt (interrupt 6), processing is aborted and will be retried next sweep.

Diagnostics

In addition to the diagnostics performed at start-up time, and slave station diagnostics are also available as follows:

Master Loopback Test

This test must be done with a coaxial cable connecting JPO and JPI on the HXLA card. The test transmits a word of data to the loop and verifies that the same data is received. This test can only be performed when the CSH is not in use. A test failure will be indicated in HRMB system error flags.

Slave Test

This test can verify the operation at any specific slave station by transmitting a series of test messages. The test transmits a word of data to the slave and requests that the same data be returned to the controller. If an error is detected in this test, it will be indicated in HRMB system error flags. Stations to be tested should appear as bits in STATIONS word ($F000_H$ test stations F, E, D, C).

--SECTION 5--

MAINTENANCE AND DIAGNOSTICS

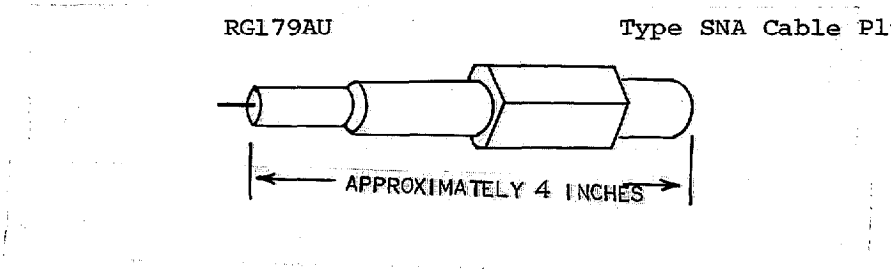
Recommended Test Equipment

* Tektronix, Model 485 oscilloscope or equivalent.

  Requirements are: Bandwidth 50 MHZ (Minimum)
  Delayed sweep cabability
  Dual channel display
  Chopped sweep capability
  External triggering capability

* Coaxial Cable Stub (Useful for observing modulation at JTO or JTI on the HLXA board.)

  Cut from spare cable to form stub as illustrated below:



* Highway Test Module

* Digital Logic Analyzer with 20 nanosecond resolution (Minimum)

* Digital Voltmeter
  Useful for measuring power supply voltages. Also can be used for measuring ohmic continuity of coaxial cables.

Preventive Maintenance

We recommend that the following preventive maintenance procedures be performed at least once each 30 days:

1. Verify that the PCMC LED (located on the HHTB board) at the head of each CSH loop, is lit. If this indicator is not lit, there has probably been an electrical break in the primary cable. This condition is reported to the operator by the PAC. Refer to the procedure for restoring primary cable operation in "Troubleshooting Symptom #1", below.

2. Verify that all threaded coaxial cable connections are at least finger tight.

Troubleshooting Procedures

Any one of the following symptoms warrant that follow-on troubleshooting procedures be implemented:

1. PCMC indicator off or intermittent when the HHTB PWB is operating in master station.

2. Failure of CSH to come up on line or remain on line.

3. Excessive error rate (less than 30 minutes) notification at the CSH terminal or PAC error typer.

Troubleshooting Symptom #1 (above)

Probable Cause. There has recently been a break in the electrical continuity of the primary cable.

Recommended Fault Finding Procedure. Perform steps 8 thru 16 of Initial Installation in Section 3 of this document.

Troubleshooting Symptom #2 (above)

Recommended Fault-Finding Procedure. Refer to this section under Diagnostics.

Troubleshooting Symptom #3

Recommended Troubleshooting Procedure. Proceed as follows:

1. Disconnect drives and devices from responding to CSH-supplied signals.

2. Use the CSH CRT terminal to exercise the problem area. The following function codes (Table 4-4) should be used from the CRT to isolate the error down to a particular I/O address: SIDI, SODI, SIEN, SOEN, GIDI, GODI, GIEN, GOEN, CLEA, INPU, OUTP. The continuous mode can be selected from the terminal to isolate faults associated with one I/O address.

3. An alternate method of troubleshooting is to use the Highway Test Module instead of the terminal, or in conjunction with the terminal to locate the faulty component. The Highway Test Module operates in any of the following modes which are discussed below:

I/O Mode/Monitor Function
I/O Mode/Simulate Function
Slave Mode - N/A
Master Mode - N/A

I/O Mode/Monitor Function--merely displays messages destined for the address dialed by the Highway Test Module operator's panel. This has limited diagnostic potential.

I/O Mode/Simulate Function--can extensively test a slave station of the DS3820 TM2, CSM or MMI variety. It should be used only when the problem cannot be located using the CSH terminal.

Repair Procedures

The only CSH repair procedure consists of either replacing malfunctioning boards or open-circuited sections of coaxial cables around the CSH loop.

Spare Parts

A Renewal Parts quotation book containing parts and material lists and catalog numbers is supplied with the Control Signal Highway. Some complete elements and assemblies may be returned to the General Electric Renewal Parts Service Center for repair and return.

A Directo-Matic® element exchange service is available for DS3800-type printed circuit cards (those used in the CSH).

All parts and repair orders should be placed with your local G.E. sales representative.

Diagnostics

The diagnostics providing for the complete I/O system checkout have been designed to test the I/O system in functional blocks. These functional blocks are:

1. Process computer interface diagnostics which are provided by the computer manufacturer to verify correct operation of its interface hardware (MPIS, DR11BD, etc.).

2. The process computer-to-HRMB RAM board data interface is completely verified every time the computer process I/O is initialized.

3. The HRMB RAM-to-HMPC microcomputer board interface is completely verified every initialize time. HMPC self diagnostics are also run every system initialize.

4. Process I/O on-line diagnostics consist of verifying correct data transmission from the process computer memory to CSH slave station I/O addresses. This on-line test verifies correct operation of the entire process I/O.

5. The CRT terminal can exercise all CSH hardware and can read and write all HRMB addresses. A complete list of all CRT commands is given in Table 4-4 of this document.

6. An off-line computer-to-HRMB interface test is provided as a troubleshooting aid (see NOTE 1).

7. The master loopback test verifies correct master station operation by isolating it from the CSH loop that it is driving.

NOTE 1: MPIS TO HRMB INTERFACE TEST

To start test, load the H4500 CS register with the test call mask (00123400) and reinitialize the computer (Alarm 100 bit 9). If the interface test is cycling, the ON-LINE LED on the HCIA board will be lit and the MPIS API echo* location bits 0-15 will contain the CS register bits 0-15. To change the data pattern, change the CS register bits 0-15 and depress the H4500 demand pushbutton. Bits 16-23 of the API echo location contains the total number of data miscompares. To stop the interface test set the CS register =-1 and depress demand pushbutton.

* (MPIS device address - 300_8) X 2 +1

If the interface test does not cycle, verify correct MPIS operation by running the Honeywell MIT module (51103047) of the System Exerciser Executive (51191050).

If the data bits 0-15 miscompare, replace master station boards HRMB or HCIA.

If HMPC board IMOK LED is not lit momentarily jumper TP1 to TP2 on HMPC (HCIA board removed). If LED remains off replace HMPC. Refer to User Guide Section 4, of this document, for IMOK LED blinking conditions.

If further troubleshooting is required use an oscilloscope and this test program to isolate the interface problem.

The following flow diagrams are provided to aid in the troubleshooting of CSH driven process I/O. These diagrams are designed to be used in conjunction with the I/O diagnostics programs provided. Diagnostics provided, that are oriented to a particular process computer, will be documented separately in this section.

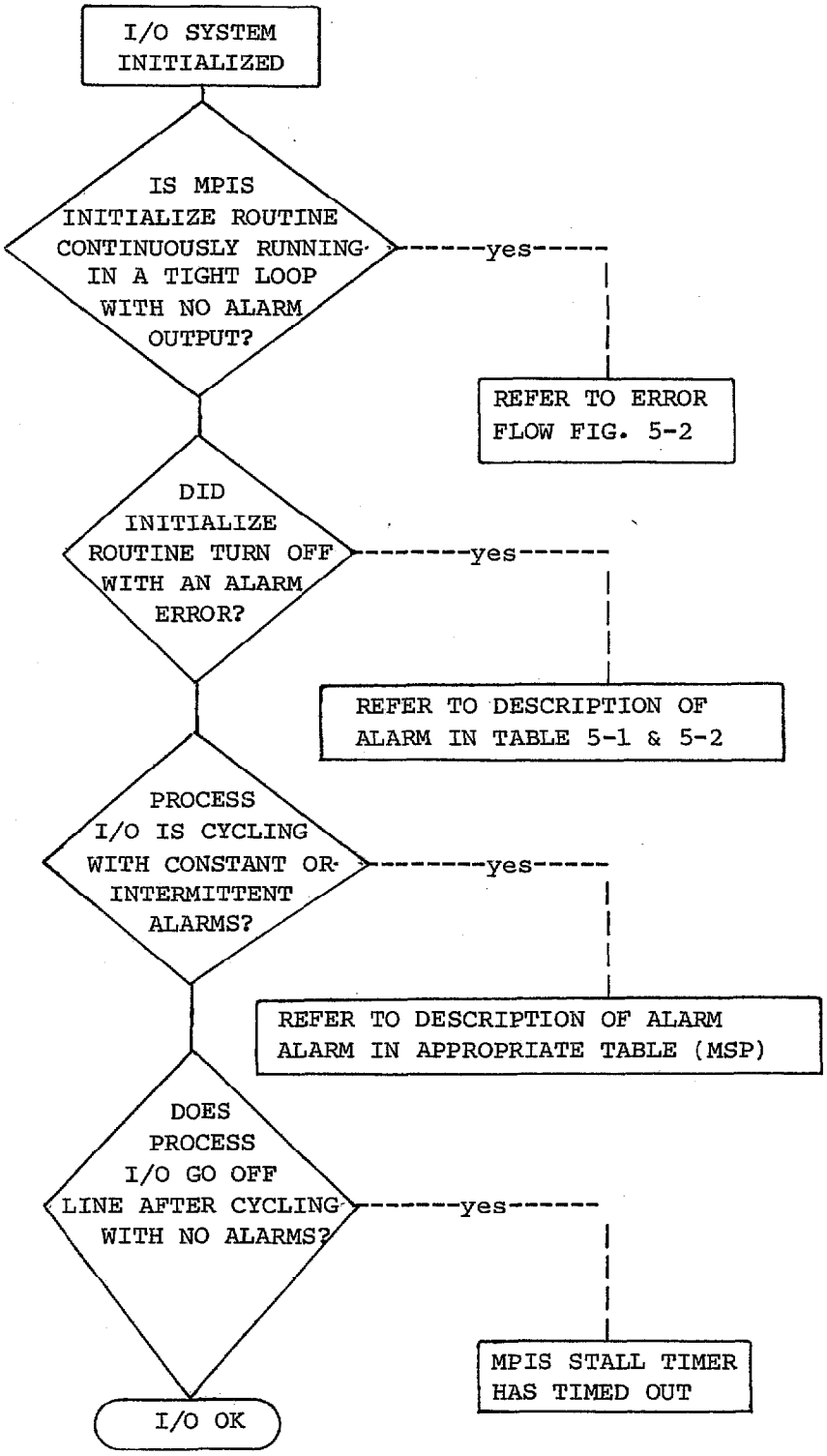


Figure 5-1
Honeywell 4500 Initialization Flow

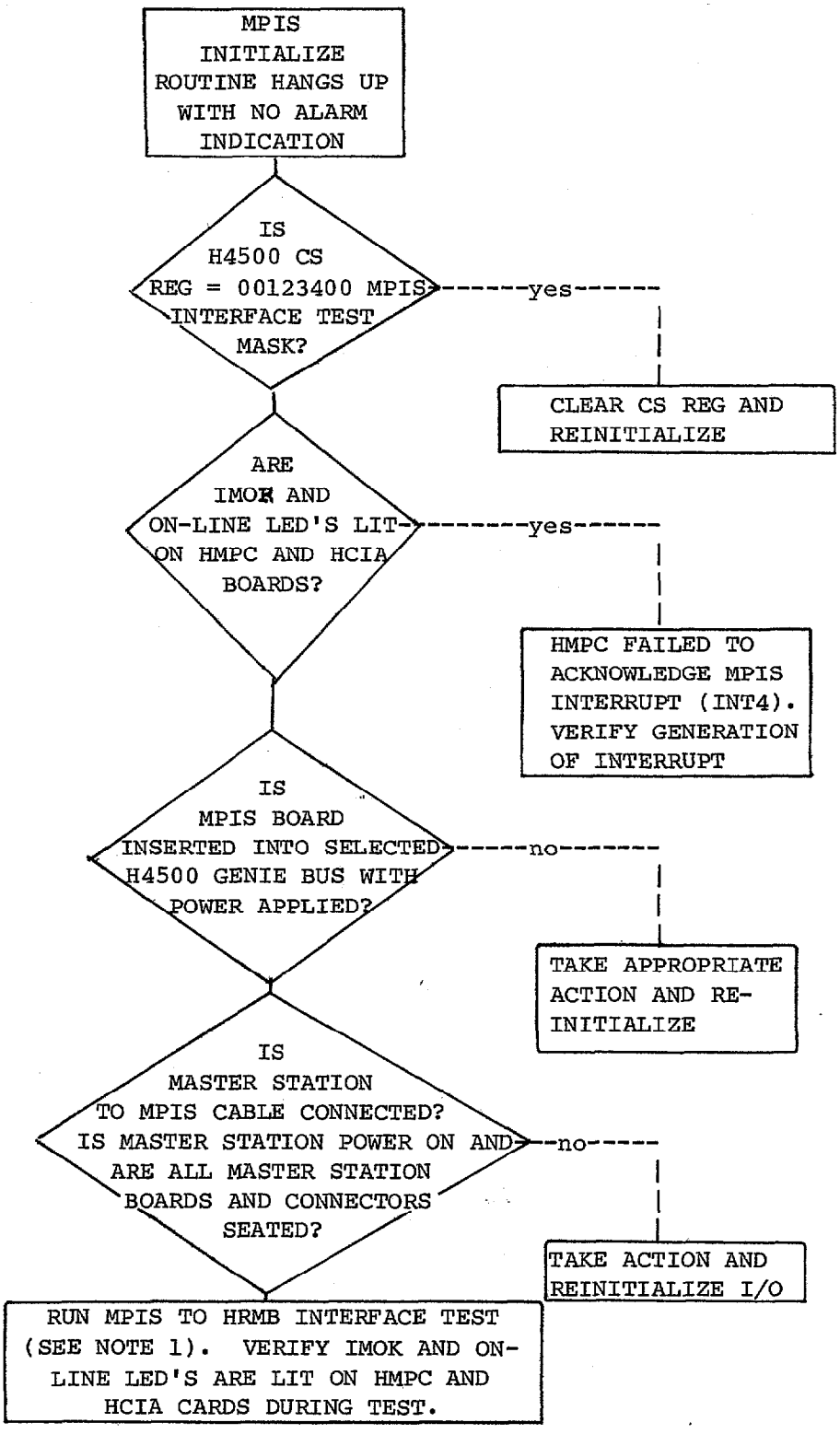


Figure 5-2
HONEYWELL 4500 Error Flow

Table 5-1
Alarm 100 - MPIS Failure

BIT SET	FAILURE	POSSIBLE CAUSE / ACTION
0-2	MPIS memory test that failed	Run MPIS diagnostics Honeywell (# 51191050, 51103047)
3	MPIS API arbitration error	"
4	MPIS DMA arbitration error	"
5	MPIS GENIE BUS parity	"
6	MPIS PROM failure	"
7	MPIS RAM failure	"
8	not used	
9	no failure mode	Indicates MPIS to HRMB interface test turned on
10	JNR instruction failure	Run MPIS diagnostics
11	JDR instruction failure	"
12	JNE instruction failure	"
13	MPIS abort failure	"
14	MPIS failed to downline load	"
15	MPIS failed to upline load	"
16	MPIS RAM load verify failure	"
17	not used	
18	HMPC stalled failure	HMPC stalled while performing function code
19	INT 4 not acknowledged by HMPC	Verify HCIA interrupt to HMPC
20	CSH disabled/master passive/ coax open	Verify status flag word on terminal and take appropriate action
21	Timeout build system para, control tables	HMPC program failure
22	Timeout scan turn on	HMPC program failure

Table 5-2
Alarm 200 - Master Station Interface Failure(H4500)

BIT SET	FAILURE	POSSIBLE CAUSE / ACTION
0	HMPC CPU error	Replace HMPC
1	HMPC RAM error	"
2	HMPC ROM checksum error	"
3	HMPC USART error	"
4	HMPC Interrupt Ctrl error	"
5	HMPC Timer error	"
6	Master Station error	Master station loopback test failed
7	Slave Station error	Test I/O function code failure
8	HMPC Bus Hog timeout error	Replace HMPC
9-15	not used	
16	MPIS-HRMB RAM verify failure	HRMB, HCIA, MPIS, cabling
17	MPIS HMPC RAM interface fail	HRMB, HMPC, HCIA, MPIS, CABLING
18	HMPC Stalled Failure	HMPC stalled during function code operation
19	INT 4 not acknowledged by HMPC	Verify HCIA interrupt to HMPC
20	Timeout on verify	HCIA board not installed or HCIA INTERRUPT TO HMPC not generated

Highway Test Module

The Highway Test Module (HTM) is a special purpose troubleshooting facility designed specifically for testing the CSH. The HTM is described in detail in GEK-65151. We highly recommend the HTM as a very effective and handy piece of test equipment.

Slave Mode Operation:

* Can originate messages at head of loop (Tests 1 thru 12).

* Can simulate missing I/O bus hand shake signals (Test 24).

* Connected to by-passed slave station (simulate activity). Can transmit to or receive from selectable I/O.

* Connected to loop (monitor activity). Can monitor message stream for selectable data-supplying or data-gathering message on the primary loop only.

* Can check validity of data and address on I/O bus of a slave station (Test 13 thru 19).

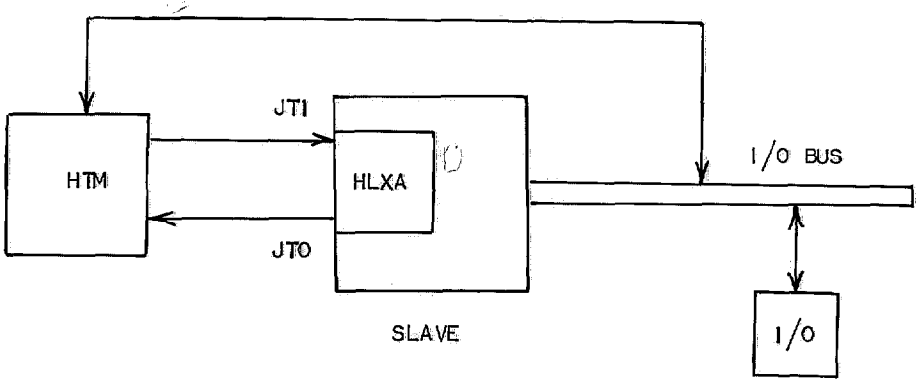


FIGURE 5-3

TEST SETUP (HTM)

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SALEM, VA. 24153

GENERAL ELECTRIC